



07: Prototyping

IN3065/INM355 | 2025–2026
User-Centred System Design

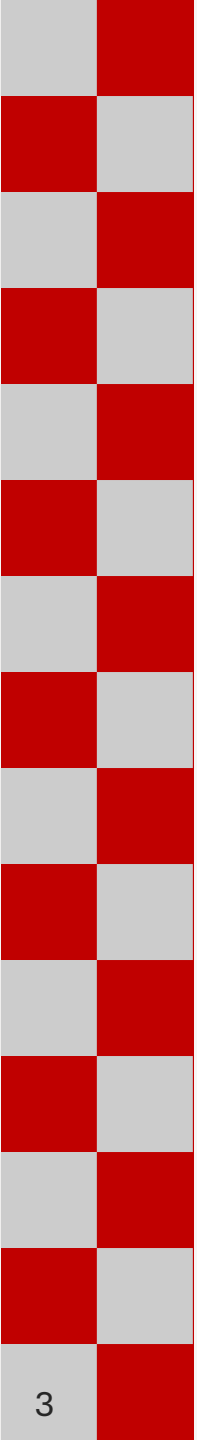
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Session 7: **Learning objectives**



By completing this session, including any practical activity and assigned readings, you should be able to:

- Explain the role of prototyping in interaction design and what prototypes may represent.
- Distinguish between key prototyping strategies (disposability, scope, and fidelity).
- Describe common prototyping techniques used in interaction design.
- Explain how task flows and wireframes help structure interaction and interfaces.



Prototyping

Understanding Prototyping

Prototyping: What is it?

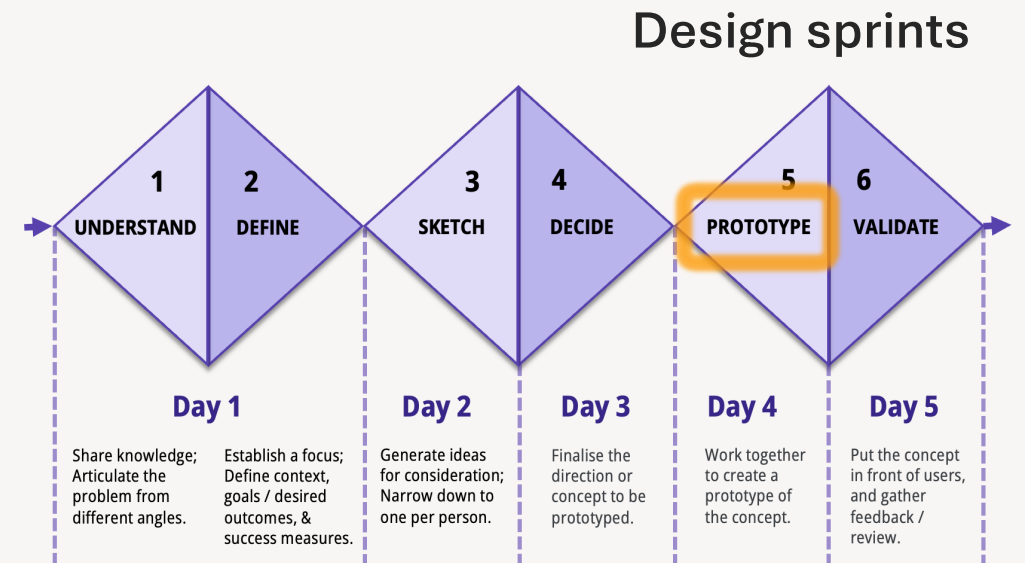
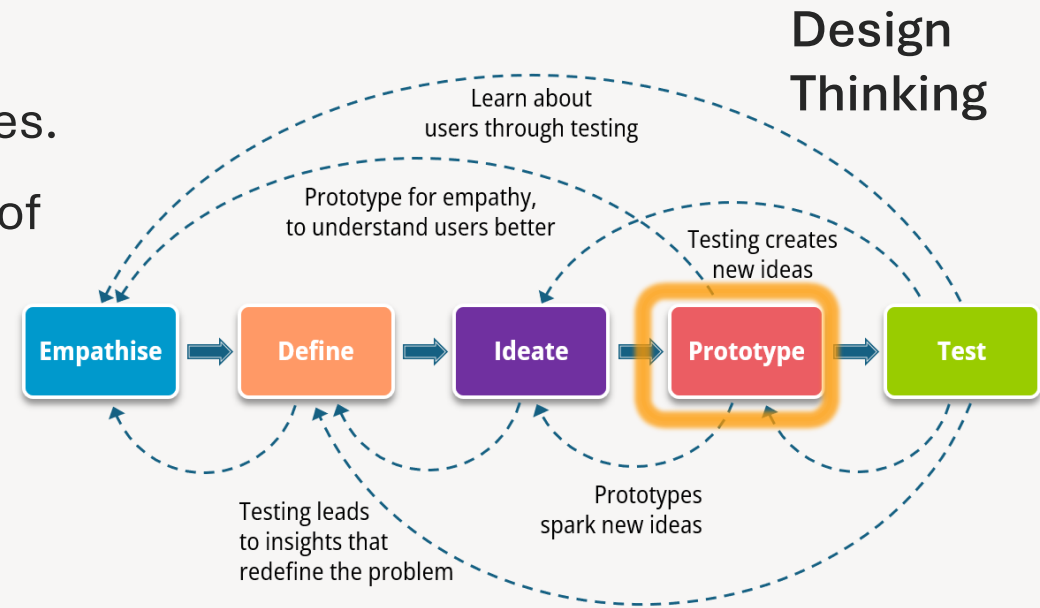
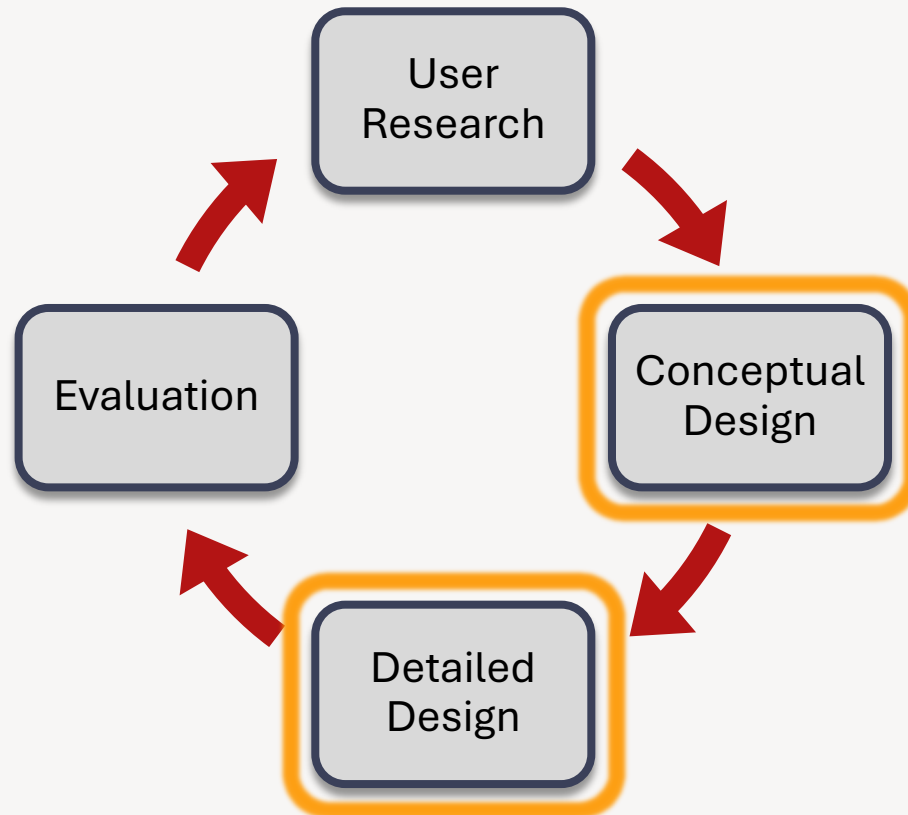
“Prototyping provides a concrete manifestation of an idea—whether it is a new product or a modification of an existing one—which allows designers to communicate their ideas and for others to try them out.”

- Sharp, Rogers, Preece (2023) Interaction Design: 12.2 Prototyping, p445
- A prototype represents (some aspects of) a design idea.
 - Not necessarily a complete representation of a product.
- Prototypes can take various forms, e.g. paper, digital, 3D prints...
- They’re used to **explore**, **communicate**, or **learn** something.
- They can be used at different stages within a design process



Prototyping within a design process

- Explicitly mentioned in some frameworks / processes.
- But, also done in others anyway, at different stages of design.



Prototyping: **Many** uses

- Formative evaluation of design, including at different stages.
- Rapid validation of ideas
- Exploring new or different ideas and techniques.
- Stimulating creativity and ideation.
- Communication.
- Selling a product / concept.
- Research.

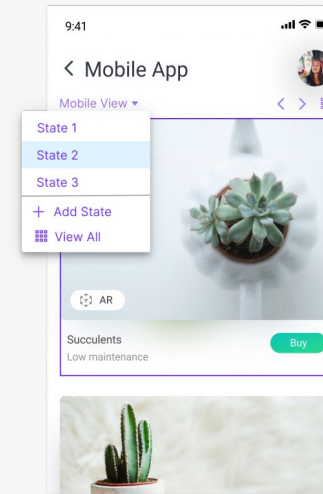
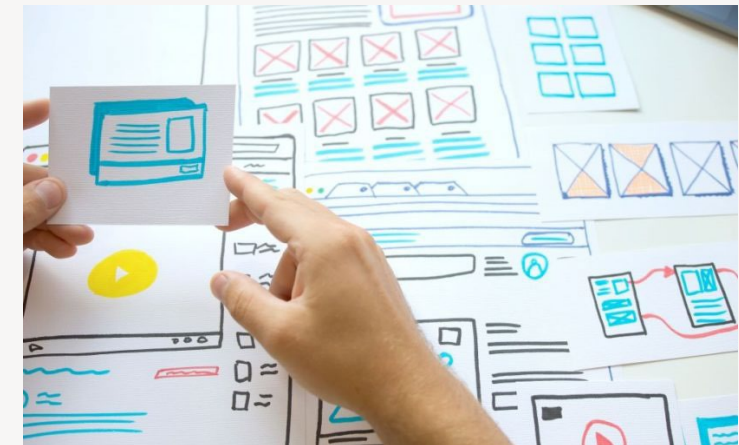


Image: Tracey Booth

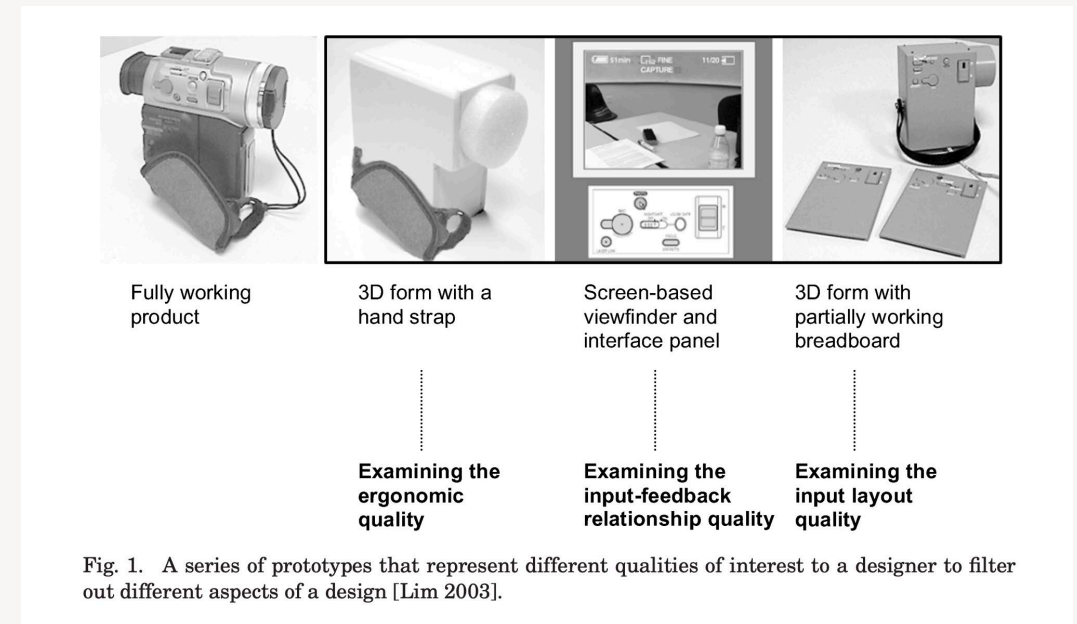


Prototypes as ‘**filters**’ and ‘**manifestations**’

- **Filter**: focuses attention on particular design aspects.
- **Manifestation**: makes ideas tangible.

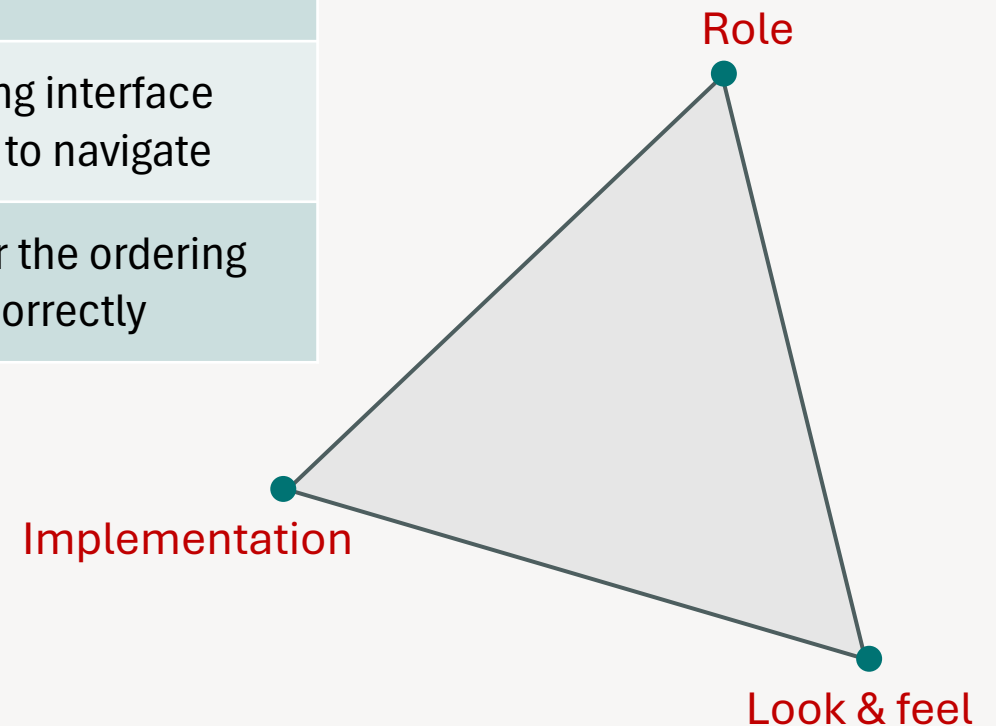
Need to balance precision vs. flexibility

- E.g. too much detail too soon can constrain creativity



What do prototypes prototype?

Dimension	Focus	Example
Role	What the system does for the user	Exploring how users order food in an app
Look & Feel	Sensory and aesthetic qualities	How the ordering interface looks and feels to navigate
Implementation	Technical function	Testing whether the ordering system works correctly



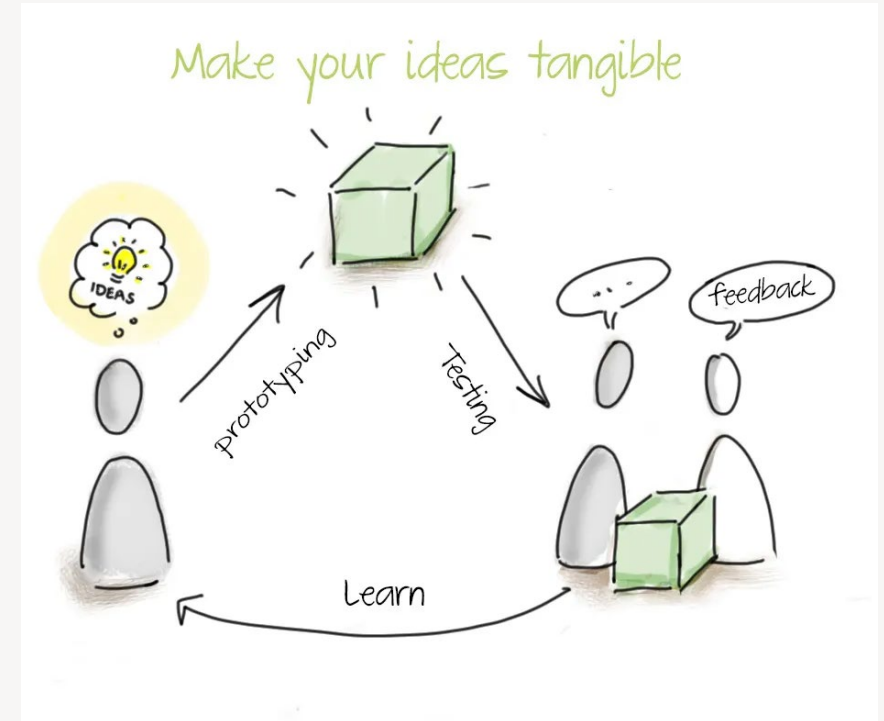
[Houde & Hill \(1997\) What do prototypes prototype?](#)

Prototyping: **Purposes**

Purpose	Explanation	Question
Exploration	To discover what an interaction could be	What might this interaction be like?
Communication	To share an idea, e.g. with team or stakeholders	How can we explain this concept?
Evaluation	To assess effectiveness, usability etc	Are there problems we might need to address?
Specification	To define how something should work before development	How should it behave?
Research / Inquiry	To generate or test knowledge through design itself.	What can we learn through making?

Prototyping: **Benefits**

- Makes abstract ideas tangible
- Encourages exploration and iteration.
- Supports team alignment and shared understanding.
- Reduces risk early in design.



*“If a picture is worth a thousand words,
a prototype is worth a thousand meetings”*

—Tom & David Kelley, IDEO

Some* strategies & techniques

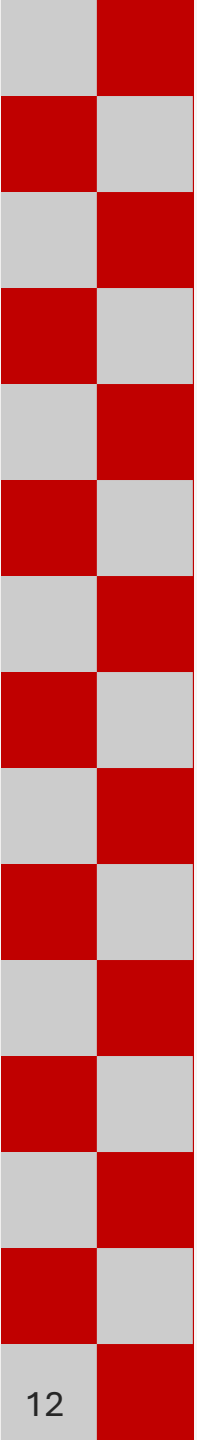
Prototyping strategies

- Throwaway vs evolutionary? (*disposability*)
- Horizontal vs vertical? (*scope*)
- Low vs mid/medium vs high *fidelity*?

Specific techniques

- Paper prototypes
- Digital prototypes
- Wizard of Oz

*There are others, but this is what we'll cover to at least some extent on this module



Structuring interaction

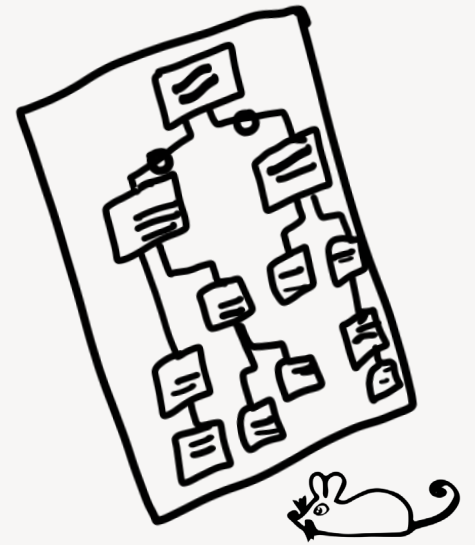
What do we prototype?

- Prototypes represent aspects of a design idea.
- In interaction design, prototypes often represent:
 - User tasks and goals
 - Flows through the system
 - Screens and interface structure
- Designers often explore these before building prototypes.

Tasks and Goals

- Users interact with systems to achieve goals.
- Goals are achieved through tasks.
- Example (CampusDine):
 - Goal: Find somewhere to eat on campus
- Tasks:
 - Browse food outlets
 - View menus
 - Choose a venue
 - Decide what to order
- Design should support these tasks clearly

Structuring interaction **Flows**



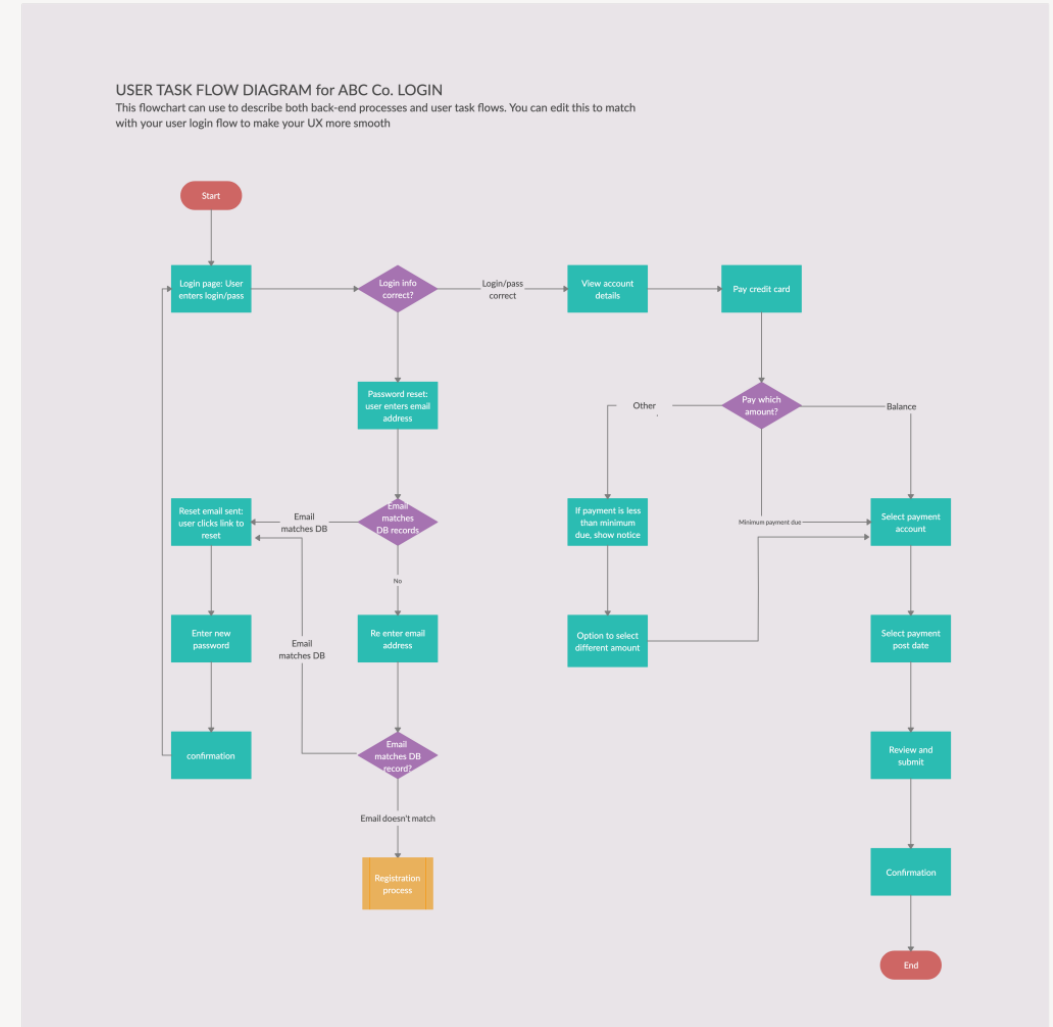
Coursework watchpoint **warning**

- User Flow diagrams are not something that you have been taught in any of part of your course.
- To gain a pass mark for this section of the coursework, you must submit the correct type of diagram.
- Please make sure you understand what you are supposed to submit for the user flow diagram deliverable.
- Pay attention to the coursework brief, including the marking scheme.



Flow diagrams

- Illustrate a user's path through a system in steps.
- Represent a user interactions with the system.
- Usually look like a flowchart.
- Difference between flows and journey maps:
<https://medium.com/productschool/the-difference-user-flows-vs-user-journeys-6546bdba13cc>



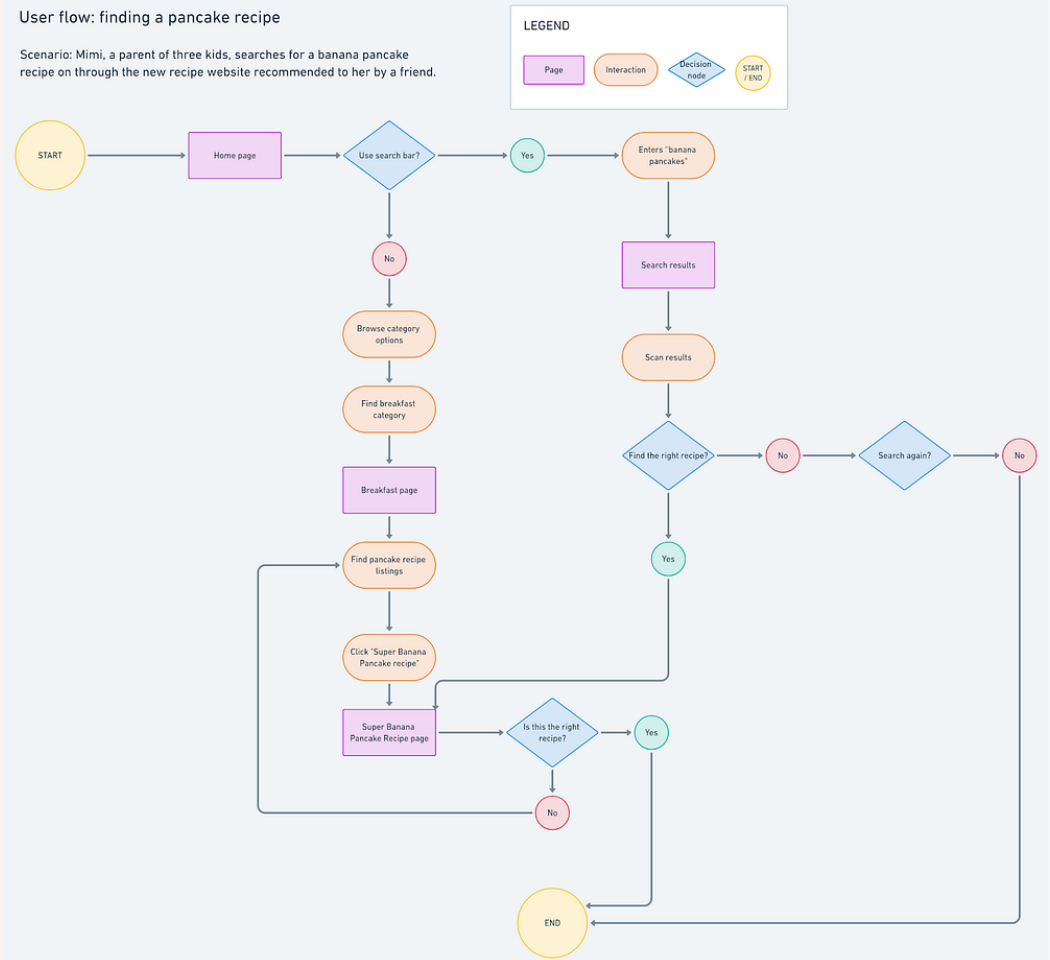
Flows: **why?**

- Prevent premature screen design.
- Provide structure for wireframes.
- Reveal branching and alternative paths.
- Ensure all edge conditions are considered.

Image source: <https://medium.com/erika-harano/ux-task-flows-versus-user-flows-as-demonstrated-by-pancakes-896e78a98026>

User flow: finding a pancake recipe

Scenario: Mimi, a parent of three kids, searches for a banana pancake recipe on through the new recipe website recommended to her by a friend.



User flows / Task flows

- Visual representation of **flow** through a system when performing a particular task—the sequence of **steps** or **actions** a user goes through to achieve a goal.
- Terms might be used interchangeably, but one way in which they are sometimes differentiated:

Task flow: Represents a single path/flow through a system;
Fairly simple sequence.

User flow: Representation considers that different users may travel different paths; Can incorporate one or more task flows; More complex, showing decision points, branches, etc.

Important: User flows are not a map of all functions or routes in a system. They are concerned with a particular task/goal.

Flow diagram elements

- Flow diagrams *typically* have:
 - A **goal** (what the user is trying to achieve)
 - **Start and end** points.
 - Series of **steps**, connected by **lines** that represent paths—arrows show **direction** 
 - **Decision points**: A choice or decision that often leads to different paths/routings.
 - **Iteration**: where part of the journey may be repeated
(be careful to avoid the user getting stuck in a loop)
- Flow diagrams *may* have
 - **Alternative paths**. Avoid dead ends.
 - **Grouping** of related steps (e.g. using shaded boxes)
 - Step labels or callouts that name pages or **cross-reference** other design artefacts, e.g. a specific wireframe ID number.
 - Further **annotations** to aid understanding

Examples: Task flow & User flow

Simple task flow

Task: Purchase a hoodie (Buy Now Checkout)

Assumptions: User searches for product and lands on product page.

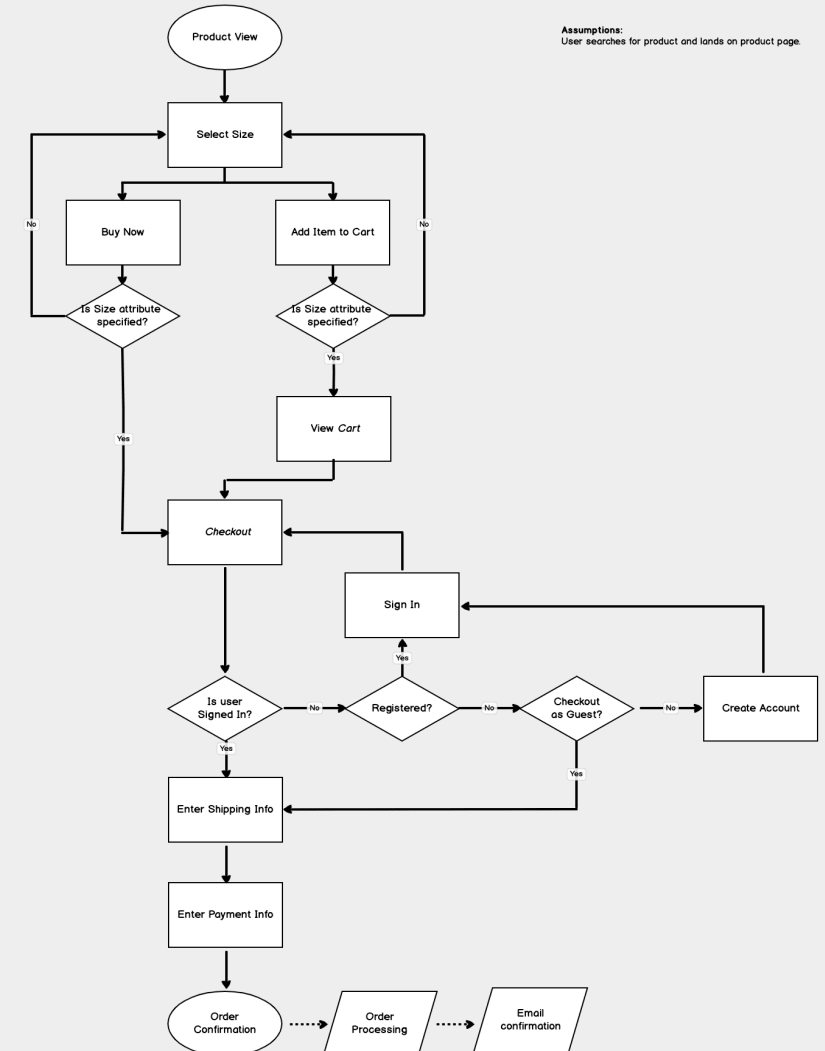


From: <https://balsamiq.com/learn/articles/wireflows>

Simple user flow

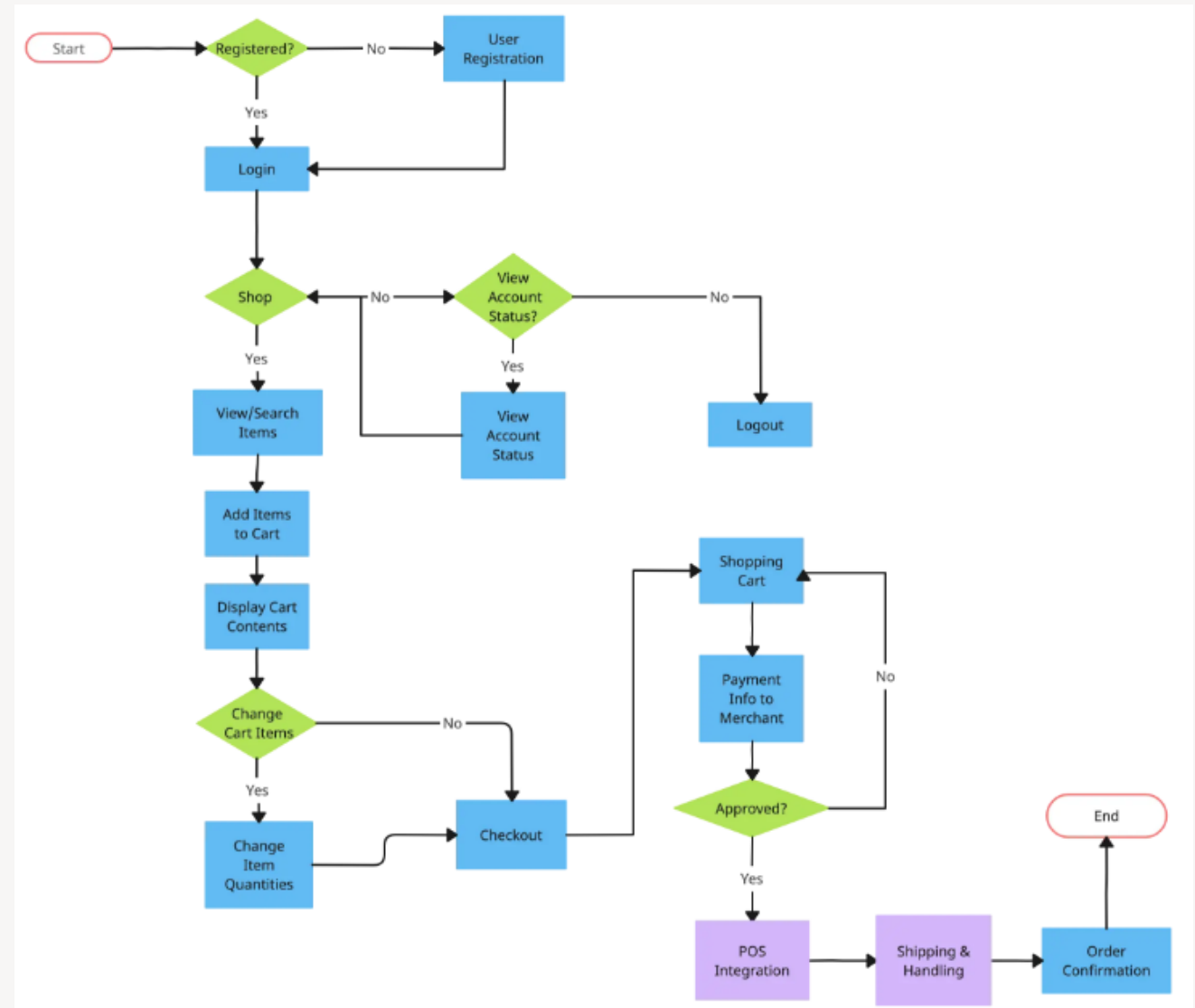
Task: Purchase a hoodie

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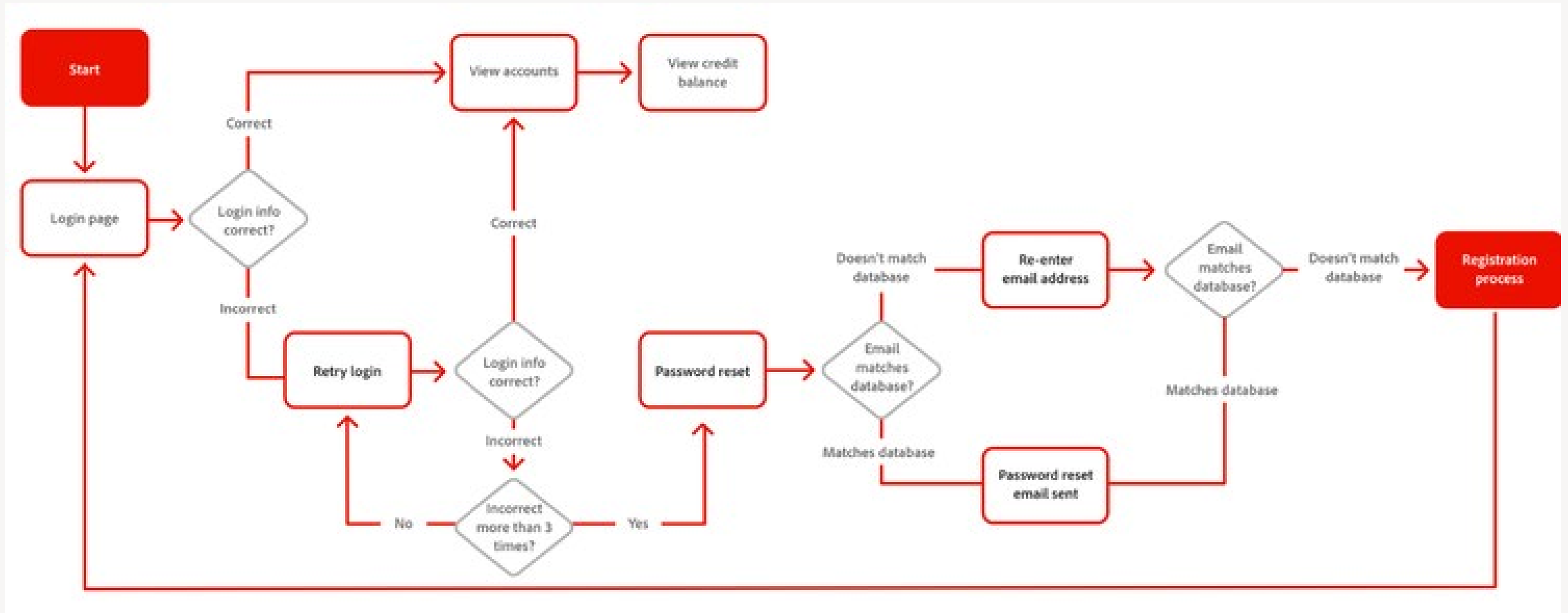


Example: Simple user flow

Note: If this was for the CW2 coursework, I would like to see the task/goal description, a key, and some indication of what screens the actions are taking place on



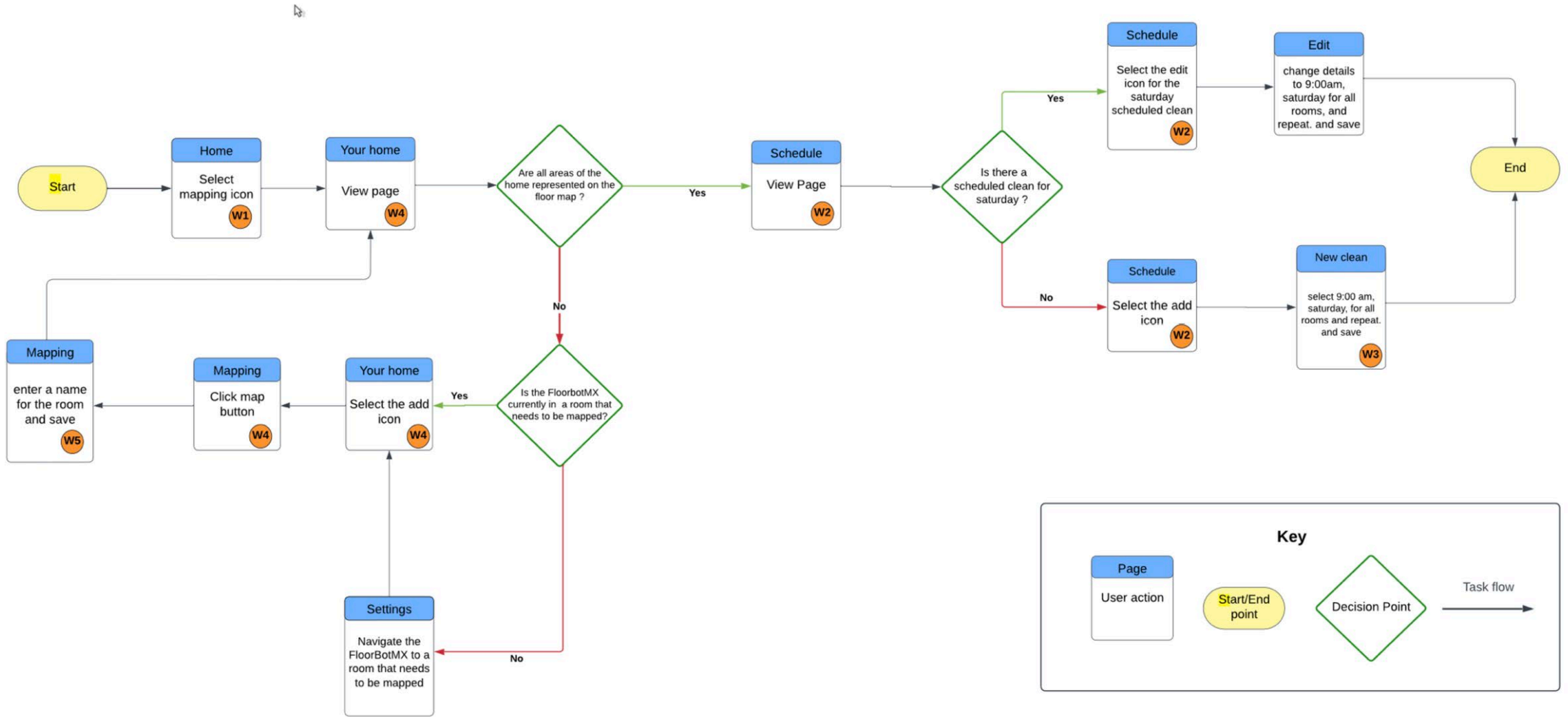
Example: Simple user flow



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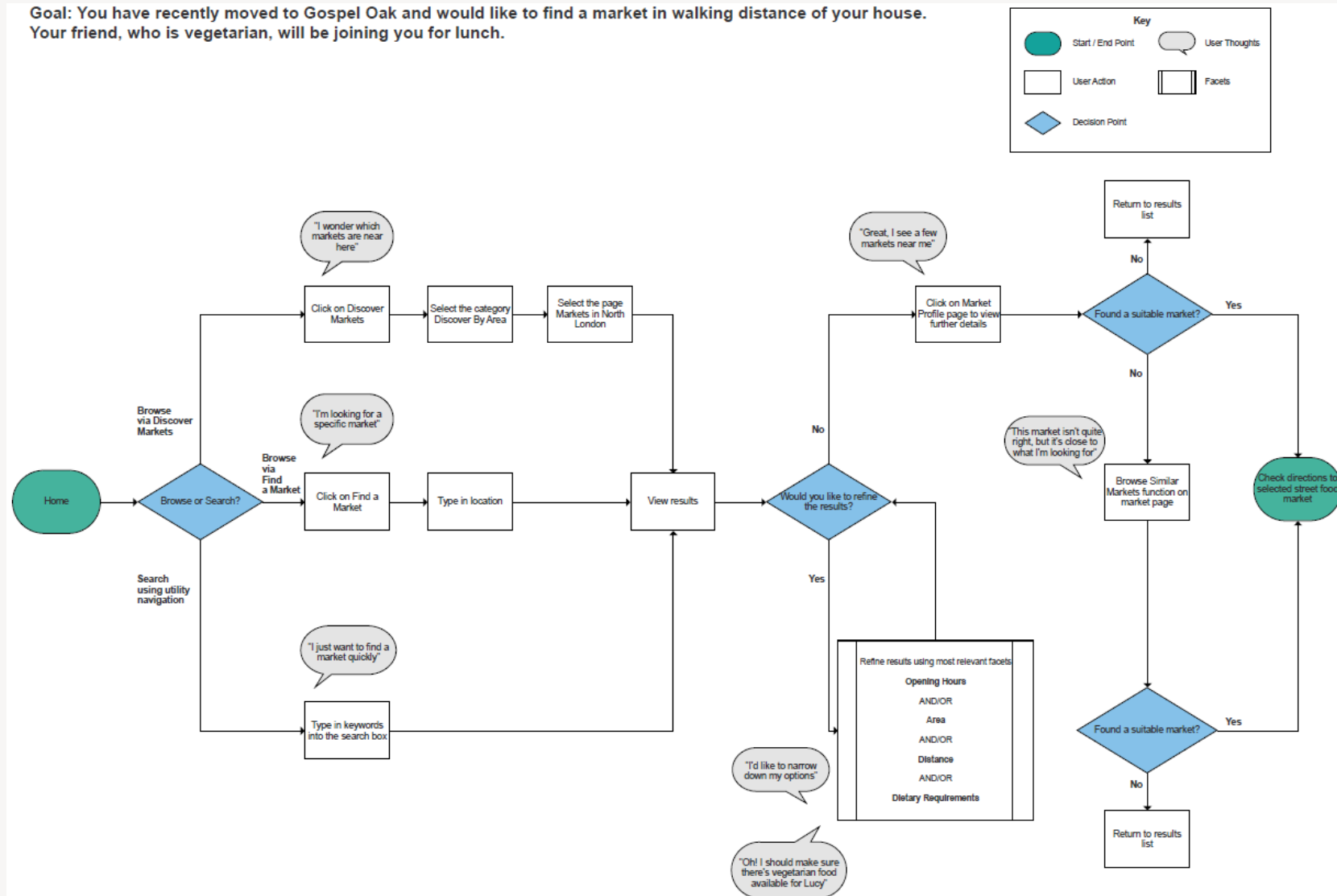
Example: User flow

Note: Can see which page/screen an action is happening on; useful cross-referencing to relevant wireframes, by using their IDs (W1 etc). The goal/task is missing though



Created by previous student

Example: User flow



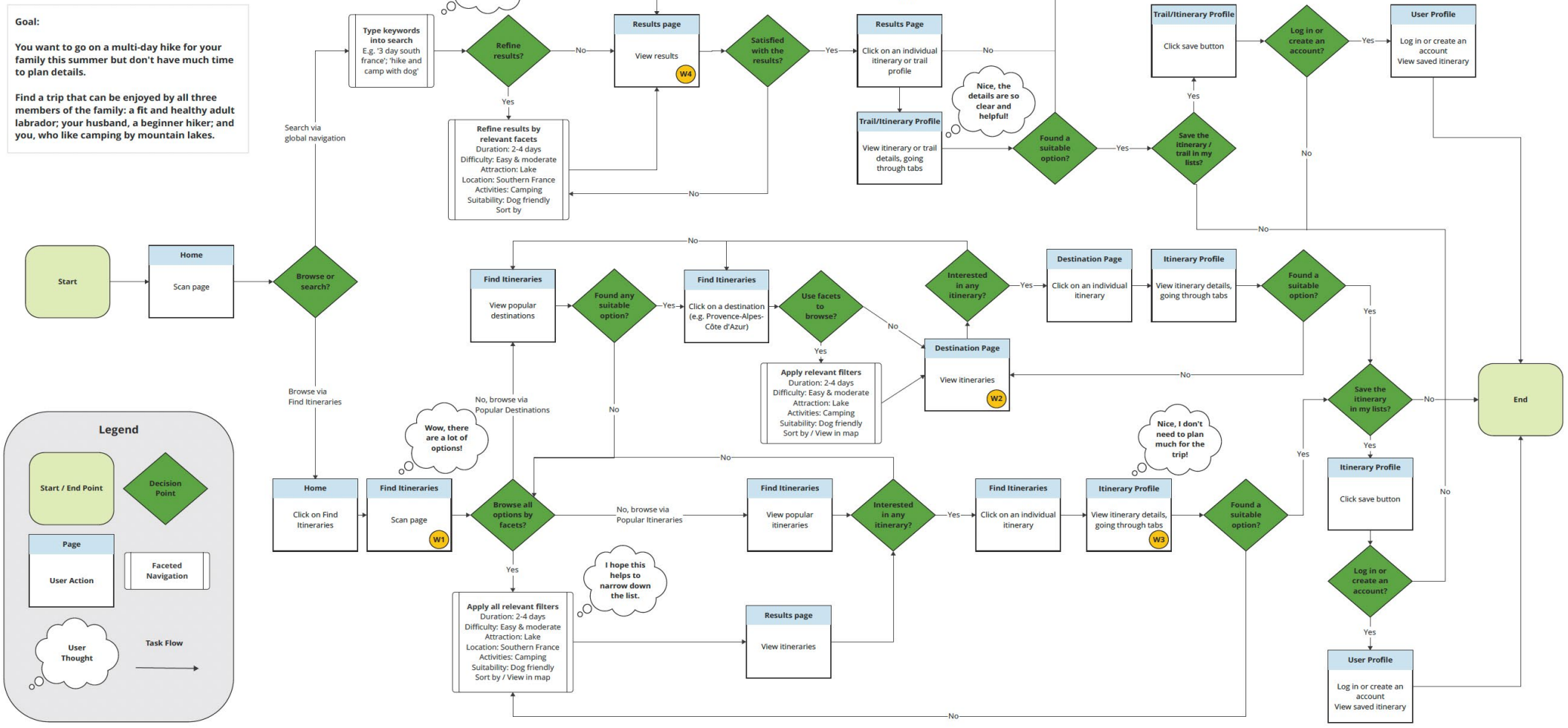
Note: this search-focused user flow diagram uses speech bubbles to provide additional useful context. There is also indication of how users are able to filter results.

Created by previous student

Example: User flow

Note: Notation choices aid the reader's understanding

Appendix 3: User journey



Example: User flow with additional design detail

User Flow Diagram

Jessica wants to modify one of her schedules:

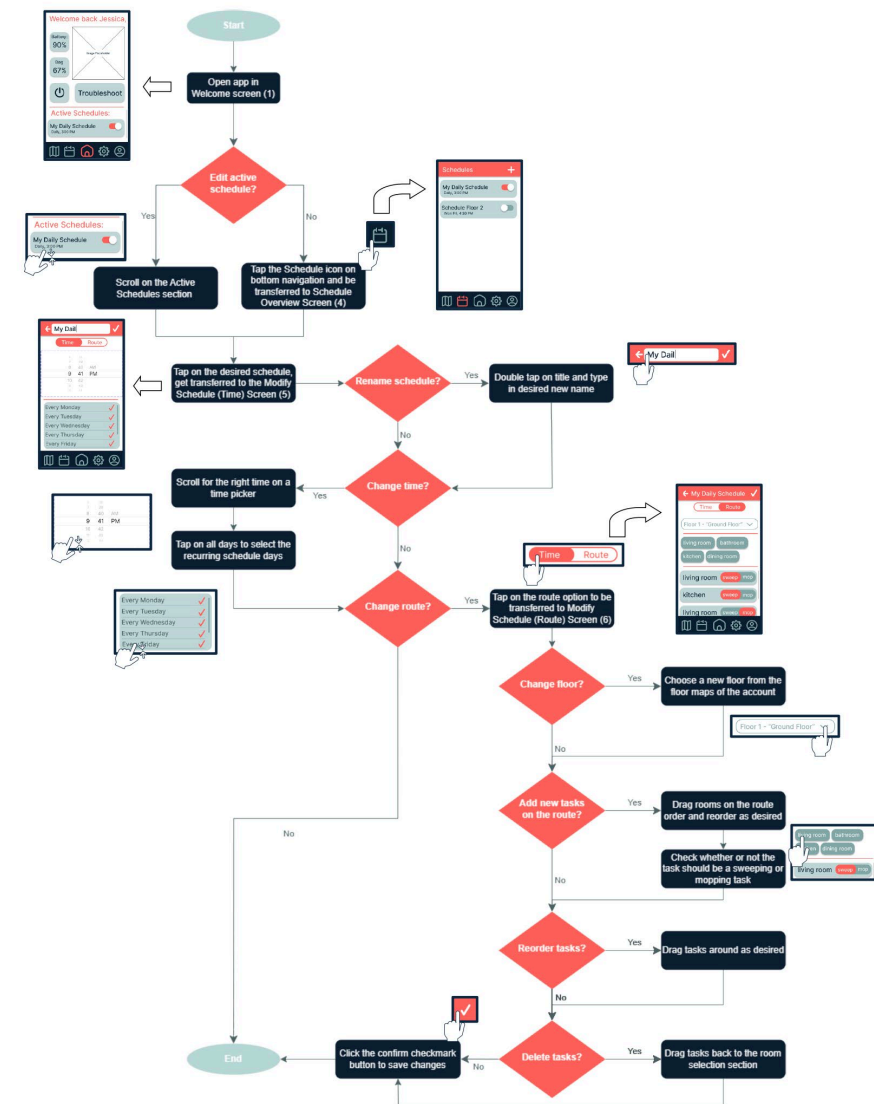


Figure 5. User Flow Diagram from Jessica wanting to modify an already set schedule, created in Draw.io.

Task/User flows within a **design process**

- Useful for determining/planning what **screens/interactions** need to be designed—e.g. before wireframing.
 - Can do this at different stages/levels, i.e. work out the basics of interaction first before elaborating.
- Also useful for spotting **gaps/problems** in a system—when planning a new system or analysing an existing one.
- **Wireflows** (covered next week) are useful for mapping out the flow of screen/page transitions and changes, triggered by interaction with specific elements—usually when you have some wireframes created.
- Useful collaboration tool—aid understanding

CW2 coursework user flow diagram

- 1 x A4 or A3 page, portrait or landscape, as appropriate.
- The diagram should provide a visual overview of the path through StudyBuddy that your primary persona will take to complete a particular task/goal—it must focus on the wireframed user journey.
- Define the task/goal at the top of the diagram.
- The diagram should have clear start and end points, and show screens, actions, flow direction, and any decision points.
- Use any software of your choice to create this diagram, e.g. Lucidchart or similar.
- This must be a user flow diagram—other diagrams will lose marks.

Marking: Quality of User Flow Diagram (10 marks)

- Quality & efficacy of flow diagram, including clarity, logic, detail, accuracy & presentation. Relationship to the wireframes.

Questions?





Structuring interaction

Wireframes

- A very brief introduction—more next week.

Wireframes

- Two-dimensional representation of an interface.
- Simplified view of what will appear on a screen/page.
- They operate as blueprints/schematics—show elements and their locations.
- Concerned with
 - Structure and layout of screens/pages,
 - Available functionality (including navigation)
 - What a user can interact with and what happens when they do.
- Usually don't contain colour/typography/images.
- Annotations can provide additional detail

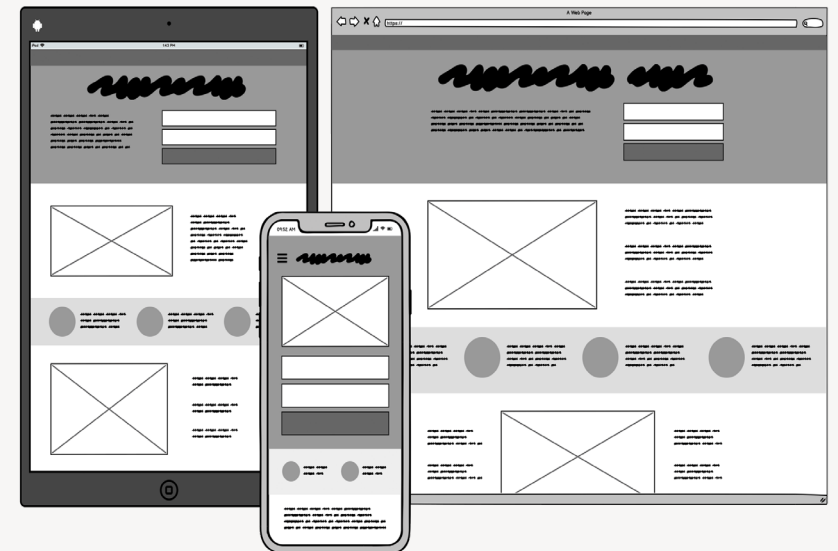
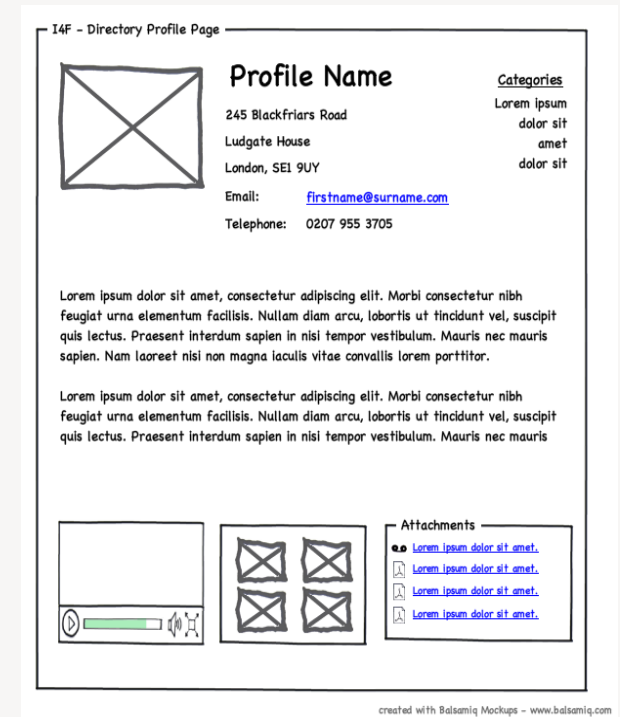
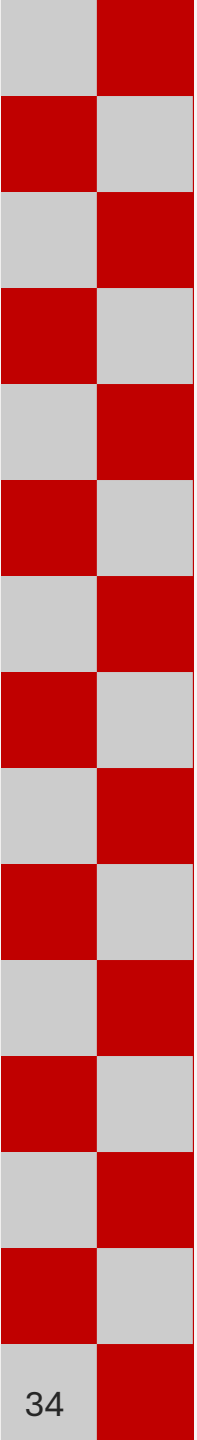


Image from: <https://balsamiq.com/learn/articles/what-are-wireframes>

Why do we **need/use** wireframes?

- Wireframes are used to:
 - Think through the structure of screens.
 - Organise interface elements and navigation.
 - Communicate design ideas to stakeholders and team members.
 - Represent screens that support interaction flows, in evaluation activities or interactive prototypes.
- In later stages of design, wireframes may become more detailed design artefacts—more about that next week





Prototyping Strategies

Consider

- **Disposability:** e.g. throwaway vs evolutionary?
- **Scope:** e.g. horizontal vs vertical?
- **Fidelity:** e.g. low vs mid/medium vs high



Prototype disposability

- E.g. throwaway vs evolutionary

Throwaway vs Evolutionary prototypes

- **Throwaway (revolutionary) prototypes**

- Built, tested and thrown away.
- Knowledge gained contributes to final system / next prototype.
- Often the best starting point when there is a high level of uncertainty, e.g. deploying a very new type of system.
- Also sometimes referred to as *revolutionary* prototypes.

- **Evolutionary prototypes**

- Not discarded but serve as basis for next iteration of the design.
- Require more rigour in development and testing.
- Danger of initially bad design persisting.



Prototype **scope**

- E.g. horizontal vs vertical

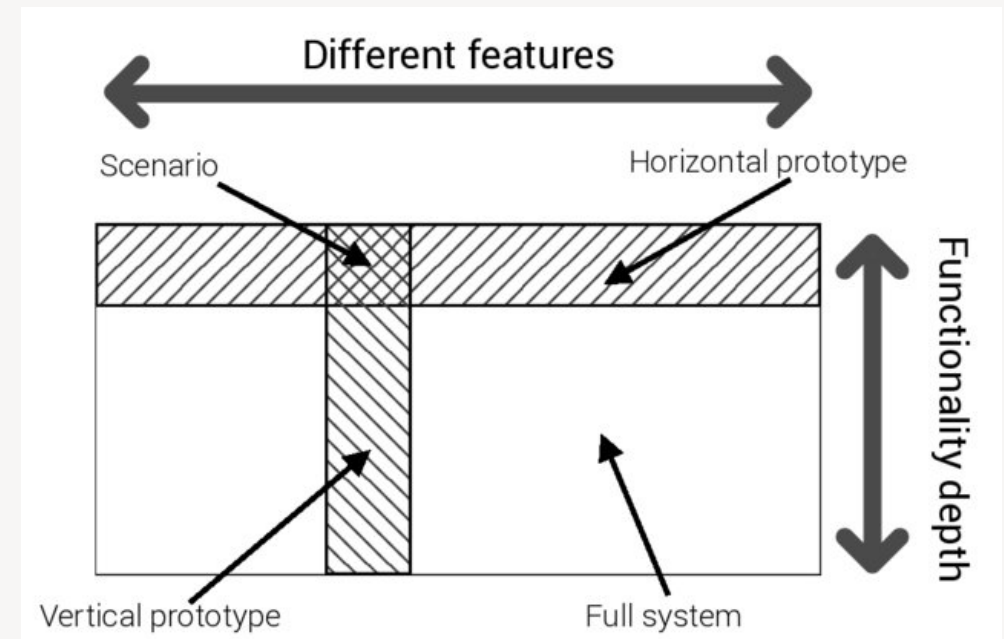
Horizontal vs Vertical prototypes

Horizontal prototypes

- Give broad coverage of features
- But less depth; superficial functionality only

Vertical prototypes

- Cover a reduced number of features
- But in greater depth
- T-prototypes can also be developed.
- Scope may also **evolve** over time





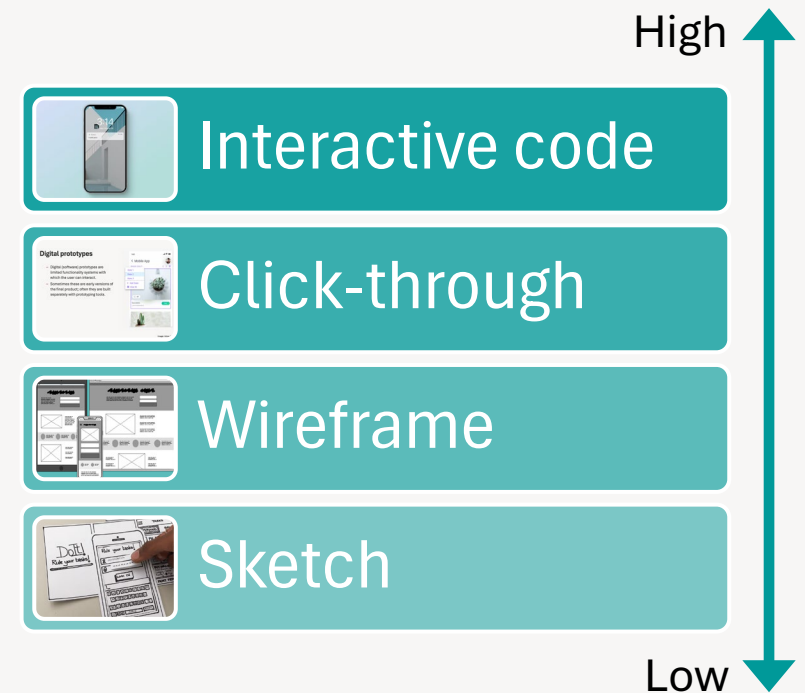
Prototype **fidelity**

- E.g. low vs medium/mid vs high

Fidelity as a **spectrum**

- **Low fidelity**: rough, quick, disposable (sketches, paper).
- **Medium fidelity**: structured, abstract (wireframes, clickable flows).
- **High fidelity**: detailed visuals, coded or interactive.

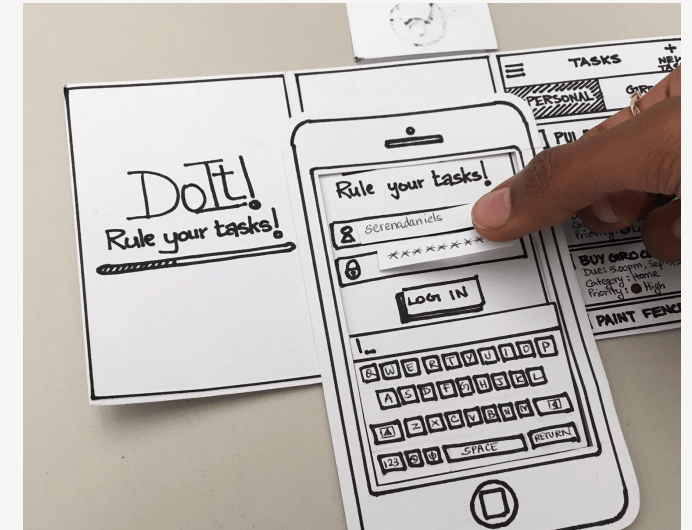
Choose fidelity to fit *purpose*, not project stage.



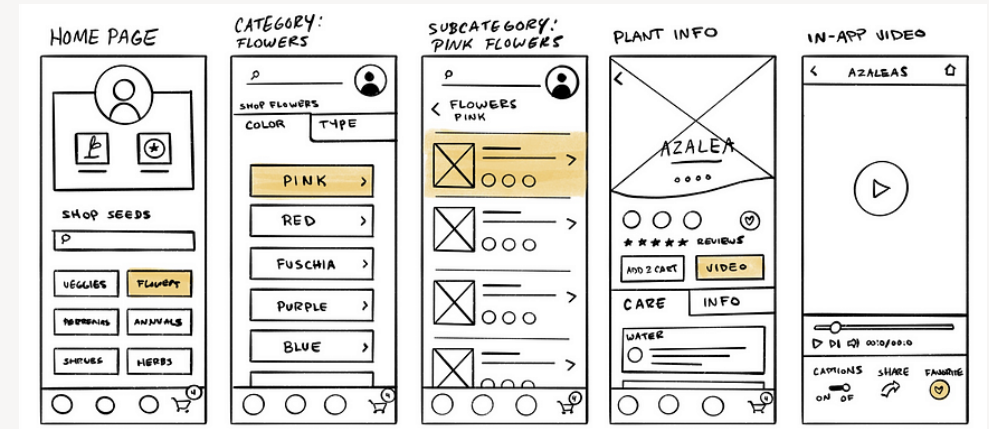
Low-fidelity prototypes

- Generally created in a different medium to the final product (e.g. paper, video, presentation software)
- Quick and cheap to create; easily changed and thrown away
- Used to test/explore concepts, and elicit user feedback as early as possible—good for weeding out major problems.
- Can *require* someone (e.g. tester or another person) to ‘play computer’ during evaluation

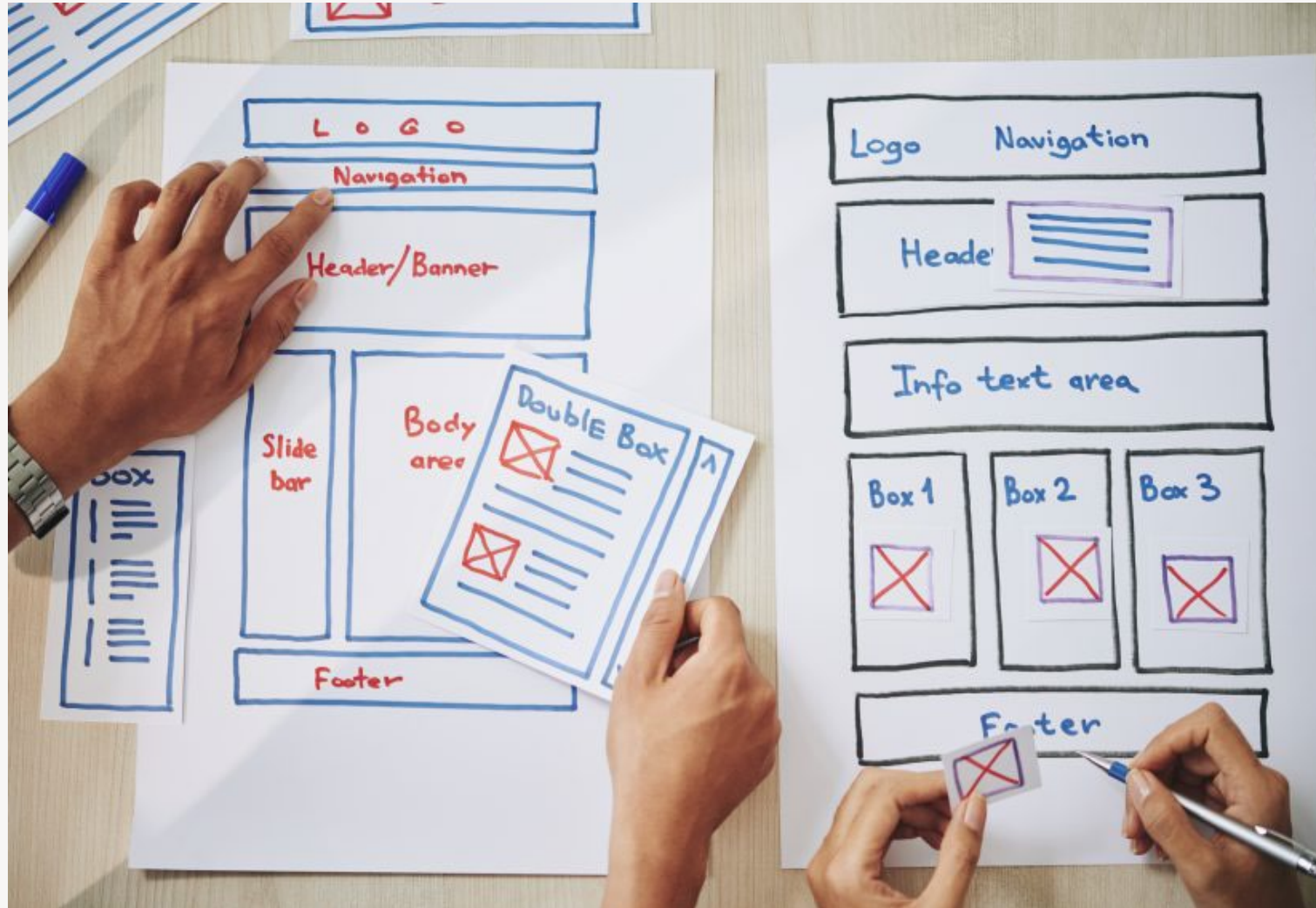
Low-fidelity paper prototype



Low-fidelity prototype



Example: Low-fidelity paper prototype

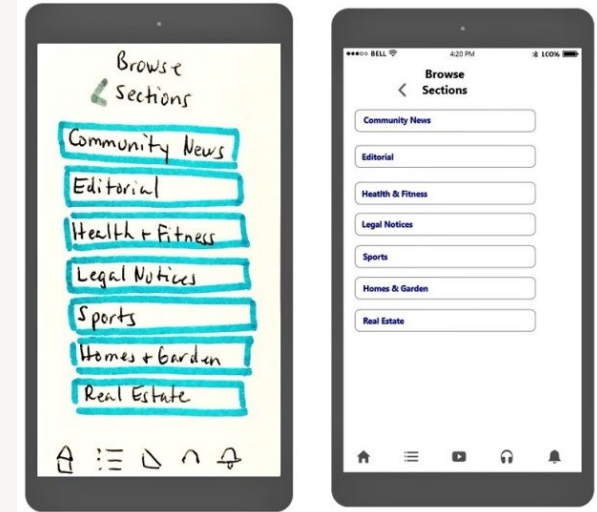


<https://xd.adobe.com/ideas/process/ui-design/what-is-prototyping>

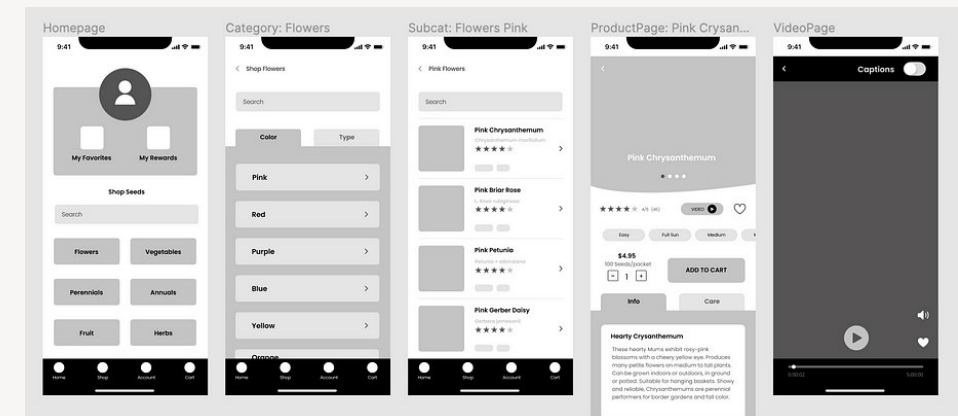
Medium/Mid-fidelity prototypes

- Between low-fidelity and high-fidelity.
- More precise representation of layout, but not realistic.
- Can be interactive / clickable.
- It used to be more time-consuming to develop in software than paper prototypes but this has made easier by current tools.

Low-fidelity vs. medium-fidelity prototype



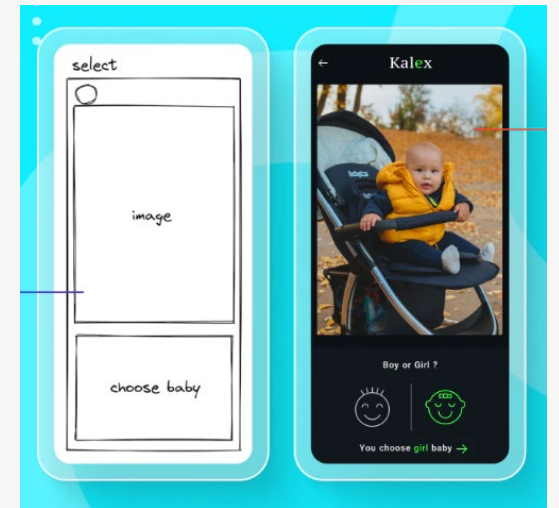
Medium-fidelity prototype



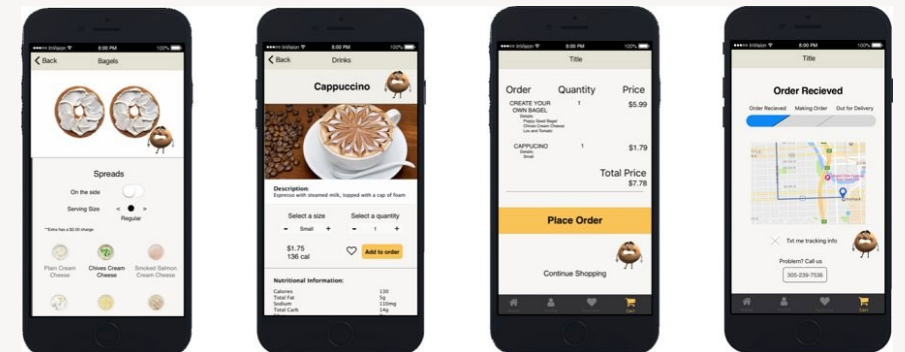
High fidelity prototypes

- More like the final product in look and feel.
- Usually created in similar medium to the final product (digital).
- May require programming
- More expensive and time-consuming to create.
- More difficult to change and discard.
- Better for demonstrating interaction.

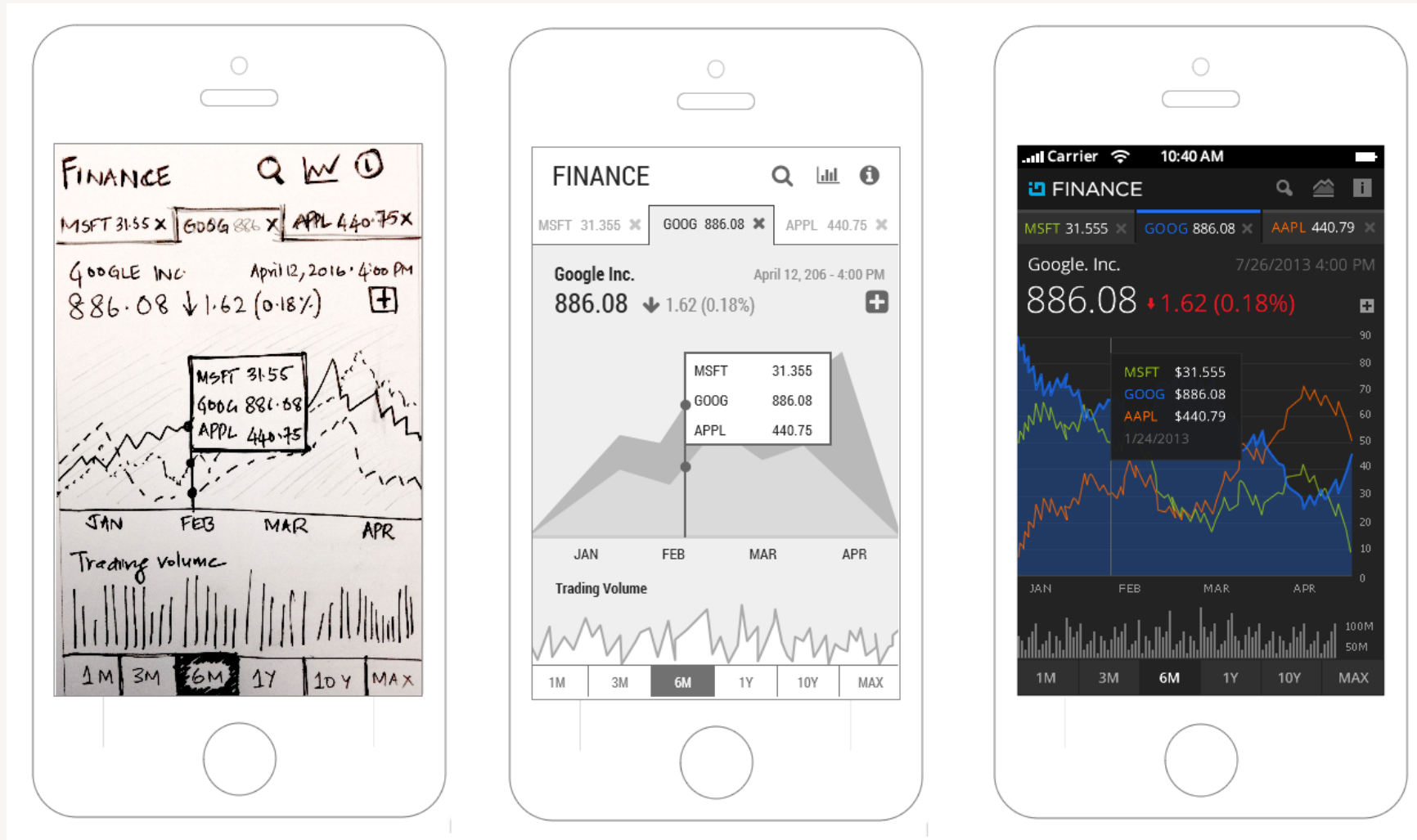
Low-fidelity vs. high-fidelity digital prototype



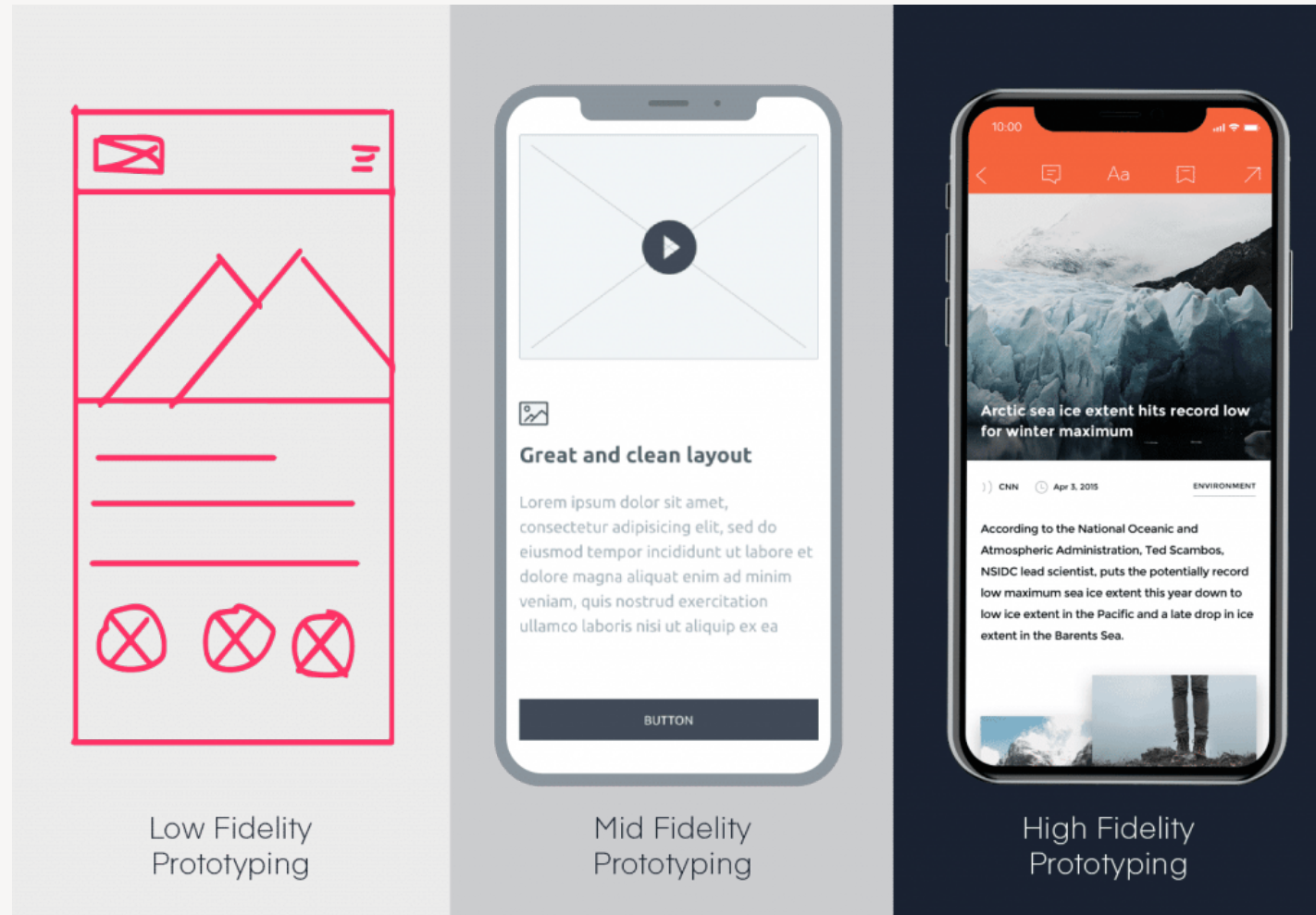
High-fidelity digital prototype



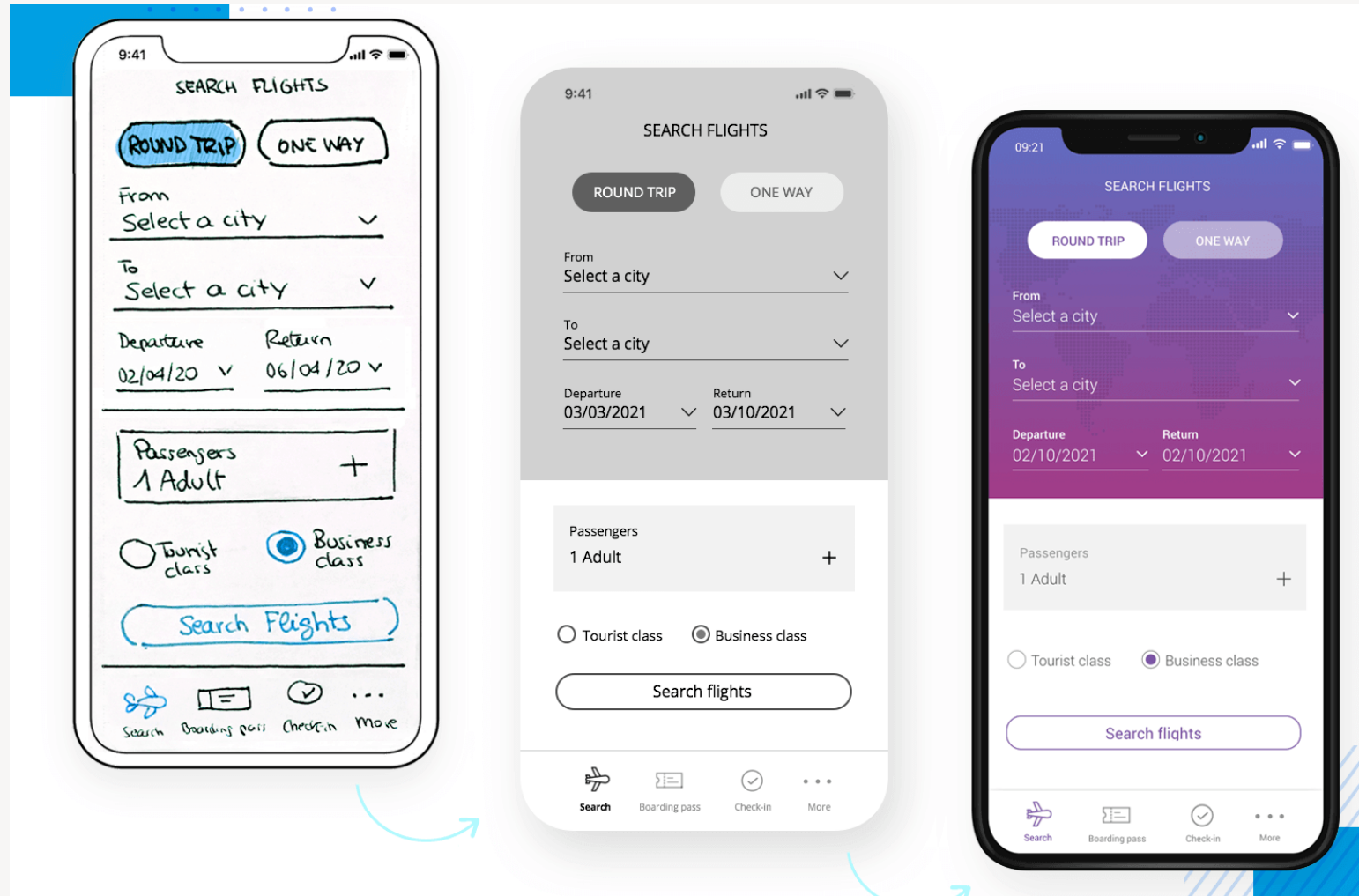
Example: Low vs medium/mid vs high-fidelity



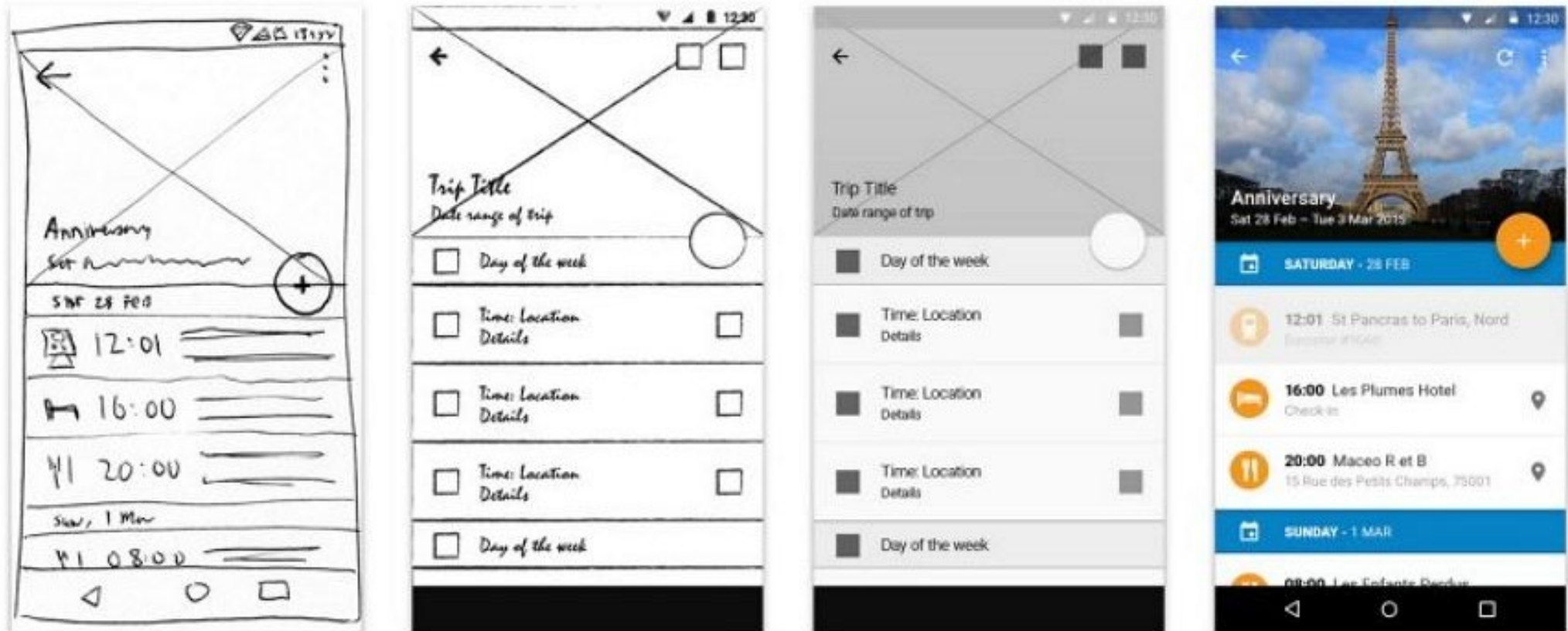
Example: Low vs medium/mid vs high-fidelity



Example: Low vs medium/mid vs high-fidelity



Example: Low vs medium/mid vs high-fidelity



So, what approach should **you** take?



- Throwaway vs evolutionary?
- Vertical vs horizontal?
- Low vs mid vs high fidelity?

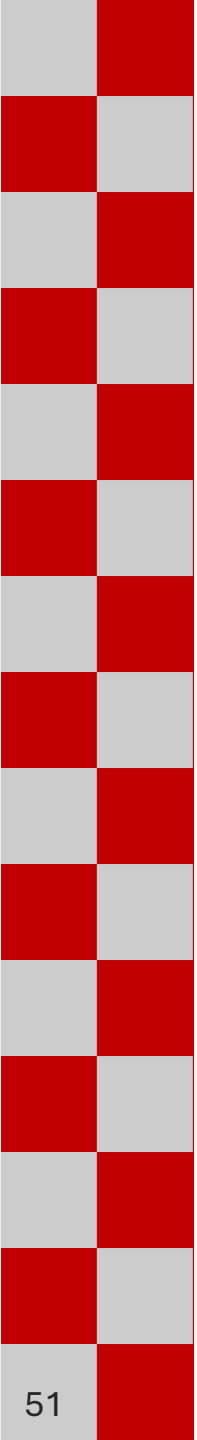
It **depends**, of course!

*“The best prototype is one that, in the **simplest and most efficient way**, makes the possibilities and limitations of a design idea visible and measurable.”*

- Lim, Stolterman, and Tenenbergh (2008) [The Anatomy of a Prototype](#)

Questions?





Prototyping

Prototyping Techniques

- Paper prototypes
- Digital prototypes
- Wizard of Oz

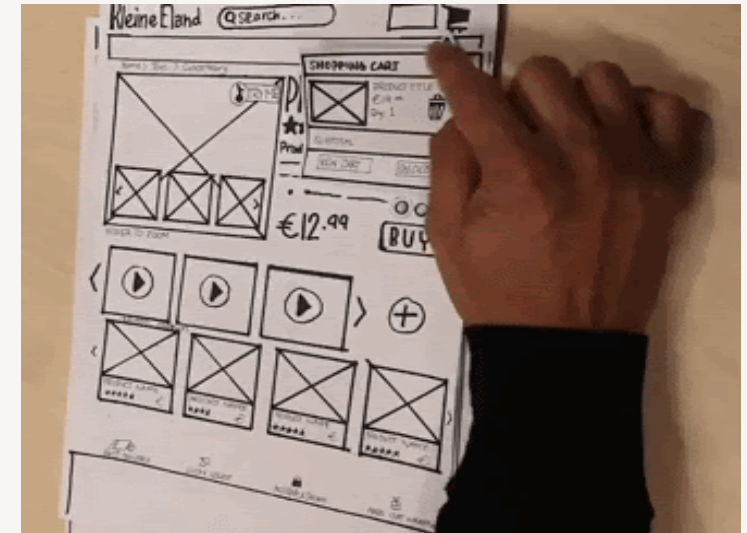


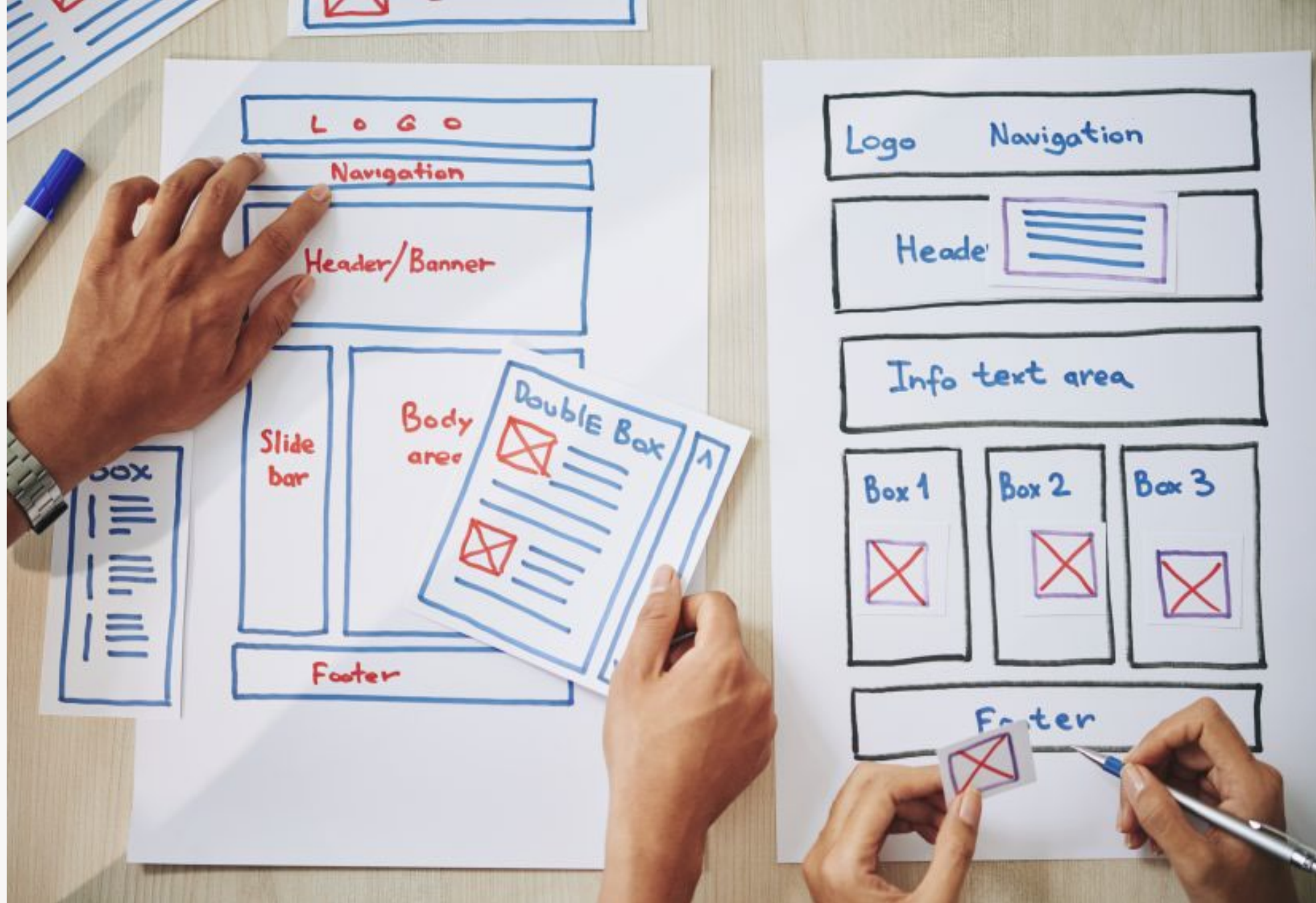
Prototyping techniques

Paper prototypes

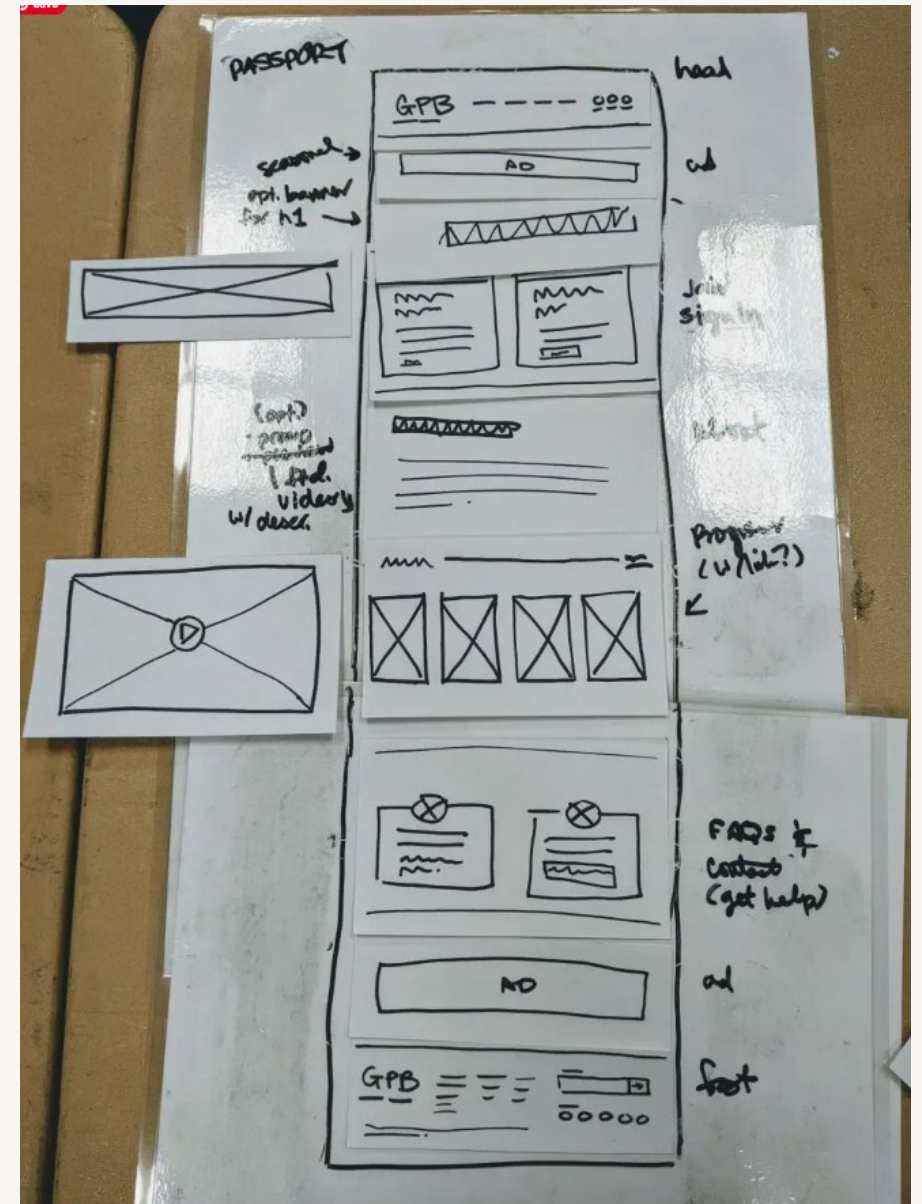
Paper prototypes

- Paper prototypes are often low-fidelity prototypes.
- Created using standard office materials (e.g. large sheets of paper, markers, acetate sheets, sticky notes, scissors, Sellotape...).
- Valuable in early exploration of design ideas.
- Supports prototyping of both physical devices and software interfaces.





Paper prototype examples



Paper prototype example

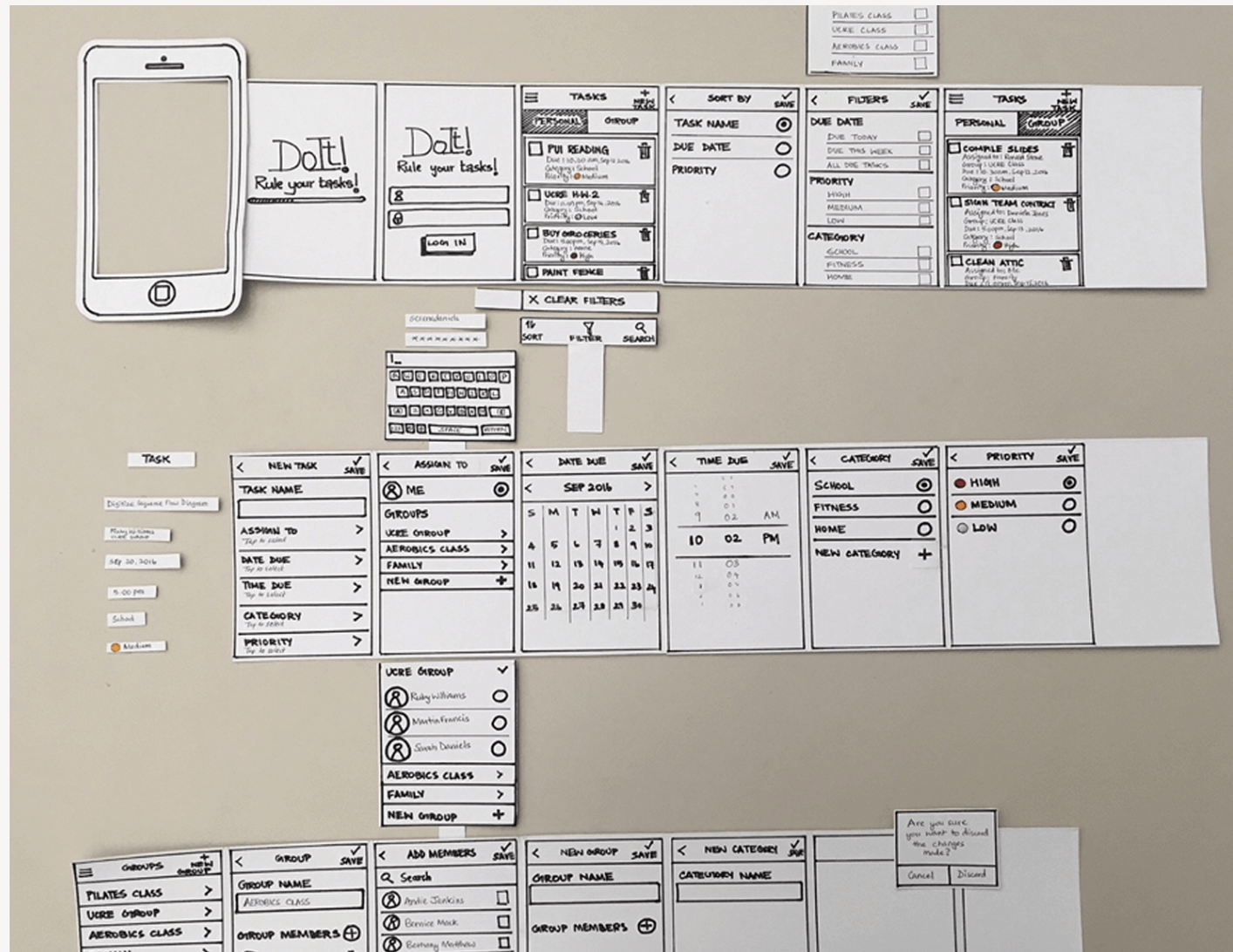


Image: <https://www.justinmind.com/prototyping/paper-prototype>

Paper prototypes in evaluation

- Test concepts and designs with users, by having someone ‘play computer’
- Purpose - to get feedback on the big things, e.g:
 - Presentation style
 - General layout
 - Terminology
 - Overall flow of interaction—major steps

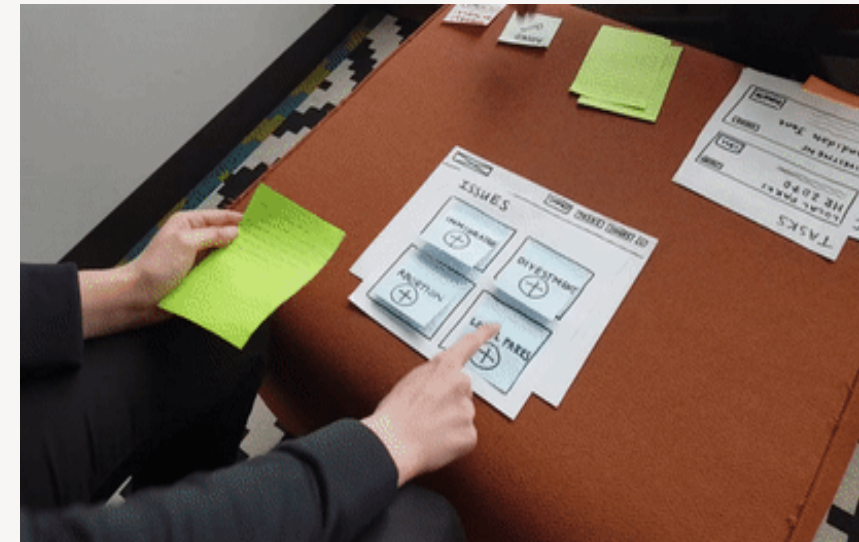
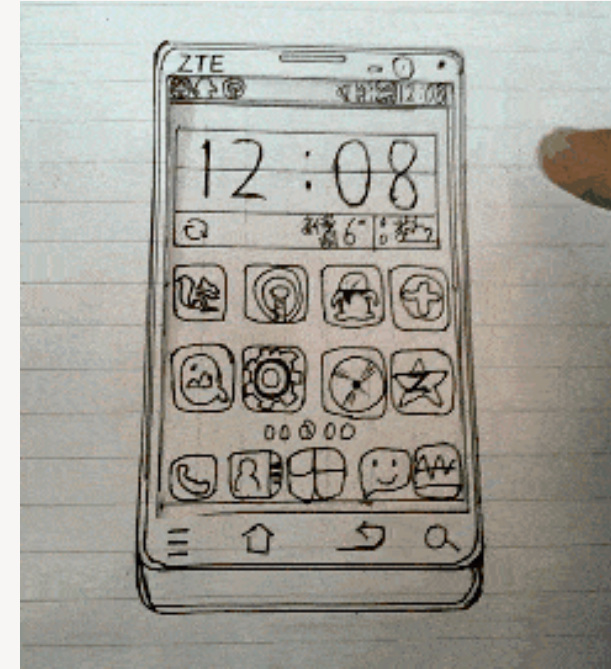


Image: <https://babich.biz/blog/the-magic-of-paper-prototyping/>

<https://www.youtube.com/watch?v=yafaGNFu8Eg>

- Tasks:
1. Open the FB app
 2. Register and Log-in
 3. Remove the home screen from your home screen
 4. Place the add \$10 Target tile to your home screen
 5. Re-arrange the tiles on your home screen

Use Notification
Indicator
Pressable Buttons



Paper prototyping: pros & cons

Pros

- Emphasis on creating the design rather than on using the tool.
- Cheap, fast to learn and use.
- Less attachment to designs.
- Good for involving users early in design and evaluation activities.
- Elicits comments on the big ideas rather than style details.
- Potential for creative designs.

Cons

- Not so good for representing detailed interactivity.
- Difficult to store, edit, manage.
- Inadequate as a standalone design specification.
- Not great for impressing clients (unless of artist-level quality)
- Design may not consider system / platform constraints & guidelines.
- Remote testing is difficult.

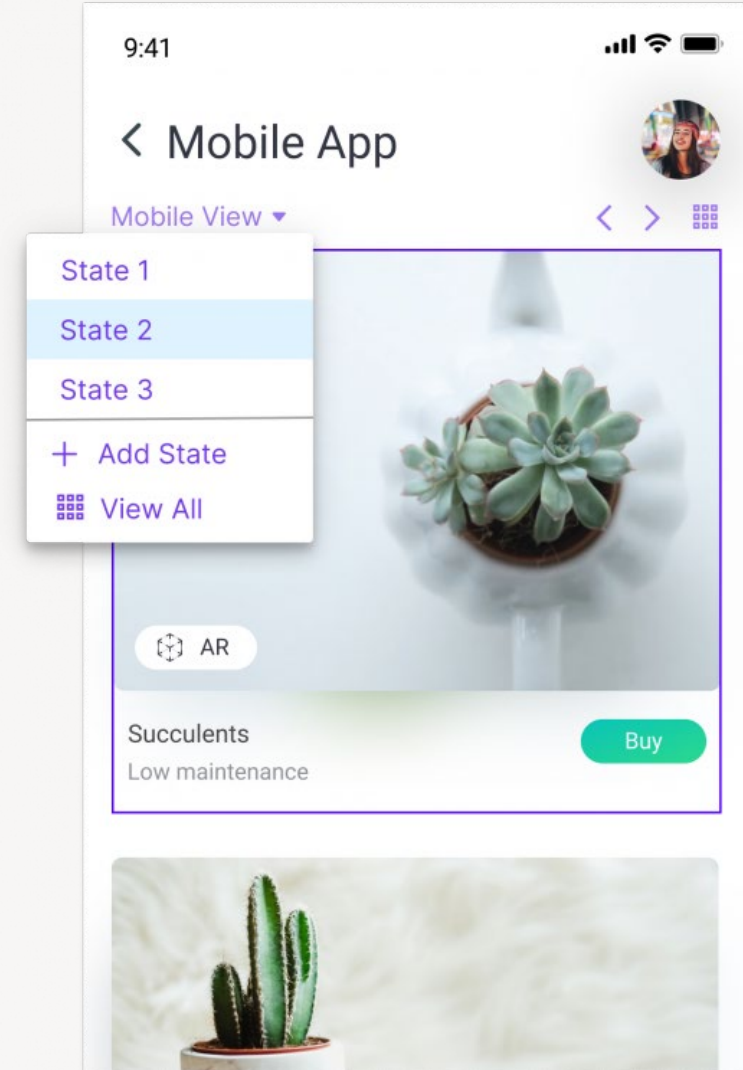


Prototyping techniques

Digital prototypes

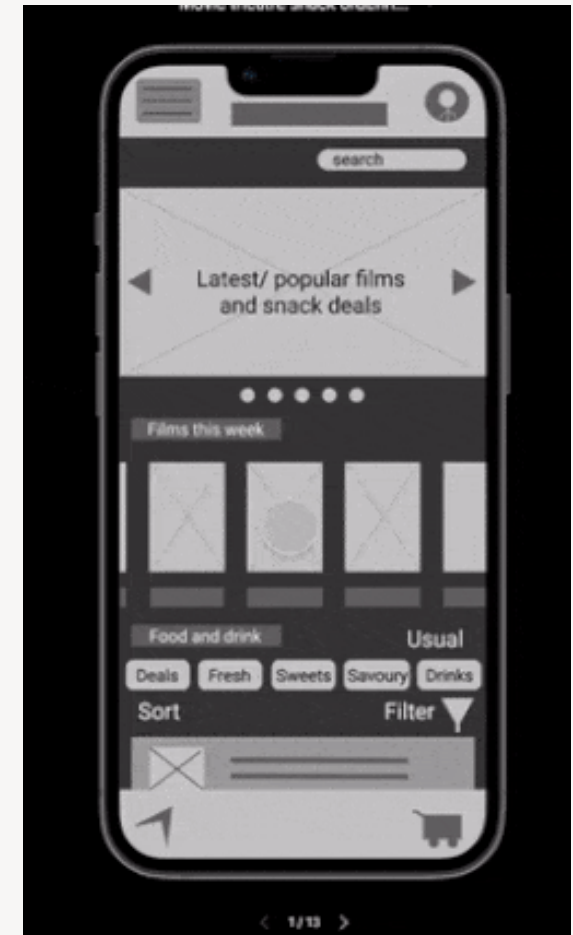
Digital prototypes

- Digital (software) prototypes are limited functionality systems with which the user can interact.
- Sometimes these are early versions of the final product; often they are built separately with prototyping tools.



Low/mid-fidelity digital prototypes

- Low/mid-fidelity interactive digital prototypes are a popular form
- Variety of tools support this activity, e.g. Figma, Axure RP, Balsamiq...
 - Some also support a progression in fidelity.
- May still require more investment than sketching and paper prototypes, but it depends...
- Recent AI-based features in tools offer new types of assistance, but can still require effort (and may detract from the main purpose)



See <http://guir.cs.berkeley.edu/projects/denim/> for further info and video of DENIM

Image: <https://www.behance.net/gallery/131744891/Cinema-snack-ordering-prototype>

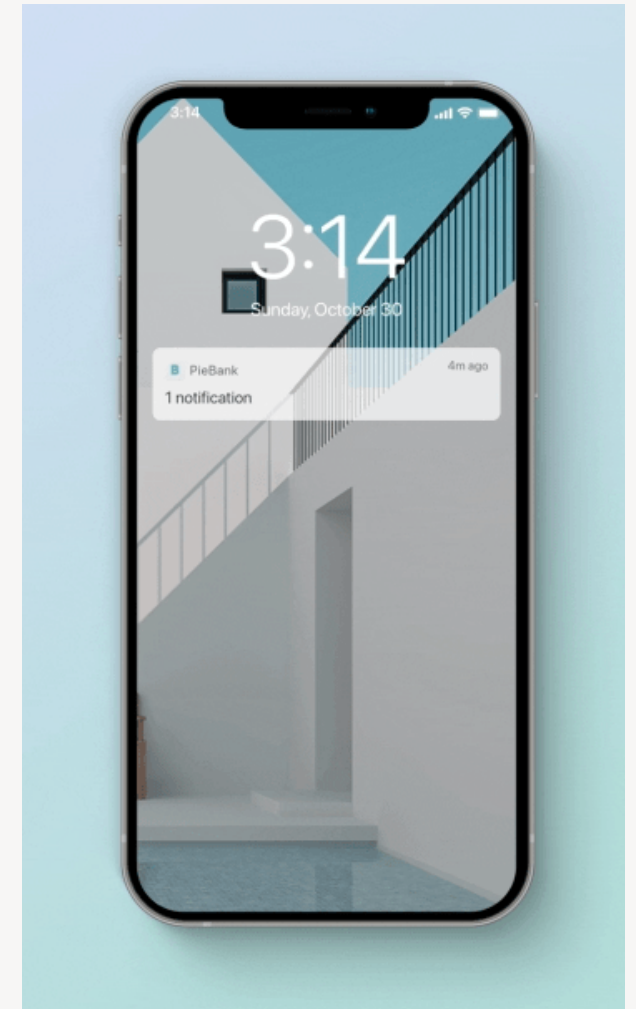
Low/mid-fidelity digital prototypes

- Allow the construction of “operational” interfaces
 - but there is no underlying functionality.
- Buttons etc. respond to user but don’t cause any calculation, invoke any functionality
 - although the screen can be made to change in response



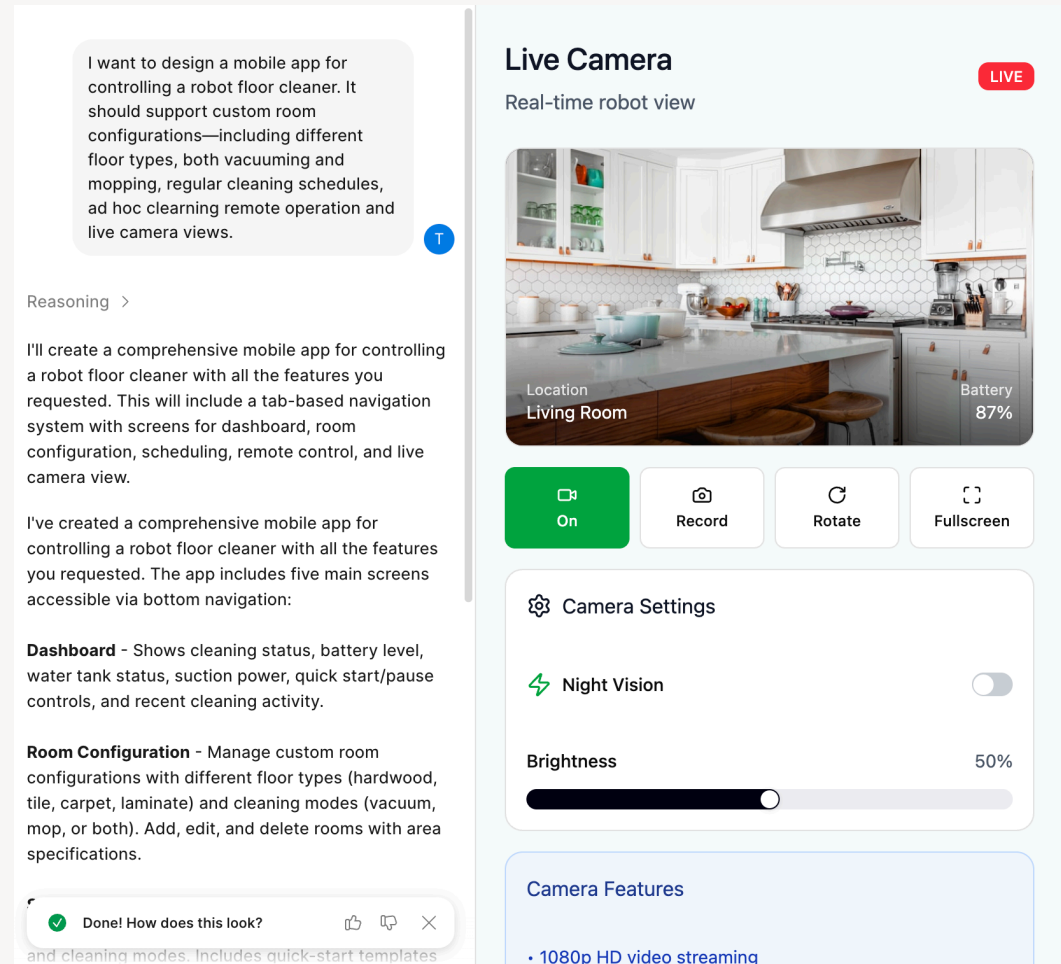
High-fidelity digital prototypes

- Usually created in more sophisticated prototyping tools.
- Look and behave more like the final product.
- Opportunity for more fine-grained interaction.
- Provide more detail and a more authentic experience.
- Useful for testing detailed design, or with clients.
- Have traditionally required more effort to produce



AI-powered prototyping

- Uses generative models to turn *ideas* into *interactive mockups* automatically.
- Designers describe intent in natural language, and the tool creates screens, layout, copy, and flow—‘co-creation’
- Useful for exploring options.
- Output can vary in quality, and human judgement is still needed—what looks good at first glance, isn’t always.
- Offer convenience, but refining can take time/effort.



Digital prototypes: pros & cons

Pros

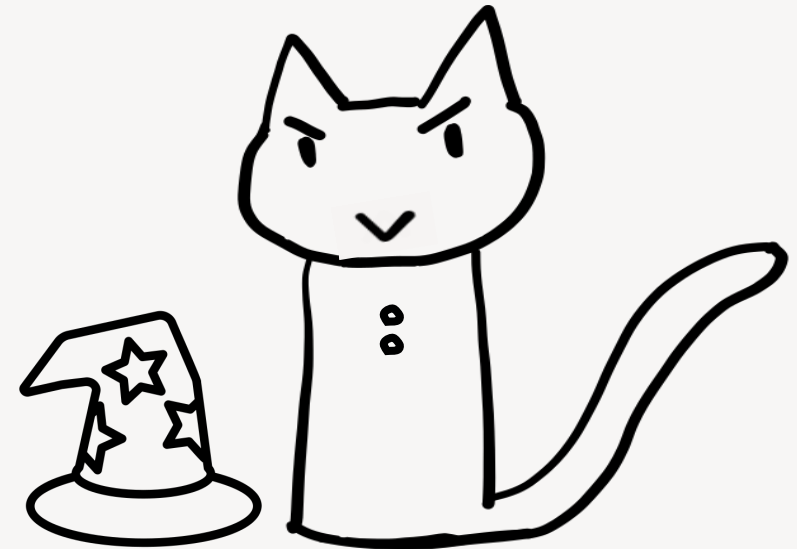
- Most reliable way of assessing the complex dynamics of a system;
- Appropriate for evaluating non-functional as well as functional aspects of systems;
- High-fidelity good for clients.
- Remote testing.

Cons

- Time & effort to develop—even with tool advancements.
- May freeze a design (it becomes difficult (or less inviting) to change);
- ‘Owned’ by the designer;
- Imposes a particular style;
- May constrain creativity.

Prototyping techniques

Wizard of Oz



Wizard-of-Oz



- Utilises human intervention to enhance a limited functionality prototype.
- Simulates system behaviour by having a human secretly perform parts of the system's function.
- The user believes they're interacting with a working system, but is unaware that a human is providing the responses..
- Can use to test interaction ideas before building complex functionality.
- Observe how users behave with “real” system responses.
- Useful for for early concept exploration or validation, service interactions, voice, AI, and chatbot interfaces
- Further reading: <https://www.nngroup.com/articles/wizard-of-oz>

Questions?



Session 7 Summary

- Prototypes make design ideas tangible, allowing designers to explore, communicate, and evaluate them.
- Prototypes may focus on different aspects of a system, such as role, look and feel, or implementation.
- Flows and wireframes help designers structure interaction and interfaces before building prototypes.
- Designers choose different prototyping strategies depending on what they want to learn (e.g. disposability, scope, fidelity).
- Different techniques can be used to create prototypes, from paper sketches to interactive digital simulations.